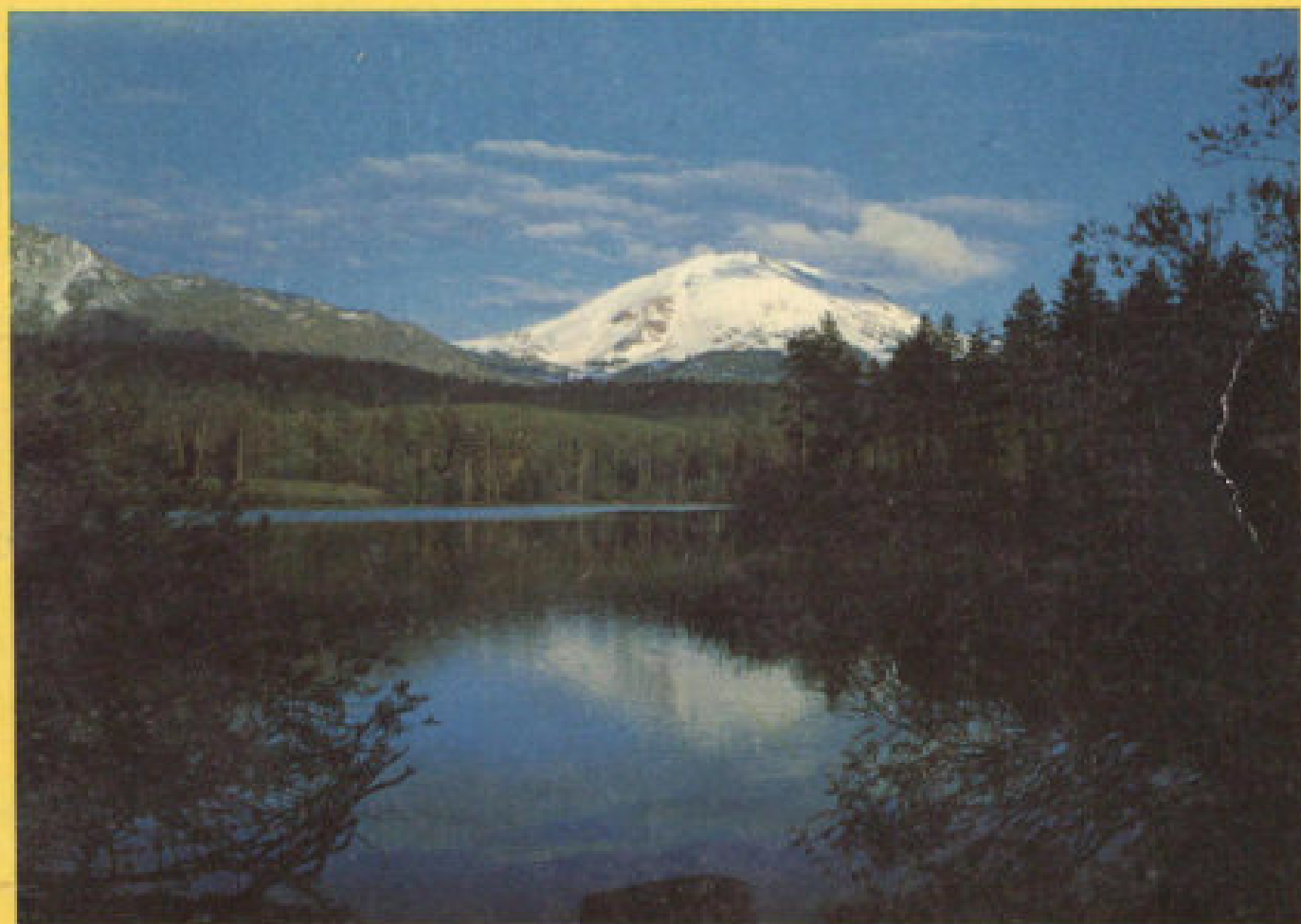


ROAD GUIDE



TO
LASSEN VOLCANIC
NATIONAL PARK

ROAD GUIDE



TO
LASSEN VOLCANIC
NATIONAL PARK

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TO LASSEN VOLCANIC NATIONAL PARK *****

The Project Gutenberg eBook, Road Guide to Lassen Volcanic National Park, by Paul E. Schulz

ROAD GUIDE TO LASSEN VOLCANIC NATIONAL PARK

by
Paul E. Schulz

LOOMIS MUSEUM ASSOCIATION



First Edition, 1950
Eighth Edition Revised 1966

COVER COLOR PHOTO
(Lassen Peak over Manzanita Lake)
by R. C. Milne

PRINTED BY LITHOGRAPHY

Lassen Litho

SUSANVILLE, CALIFORNIA

HOW TO USE THIS GUIDE

This booklet describes the more important points along the Lassen Park Road. To get the most enjoyment from this pamphlet make a leisurely trip—drive slowly. Visitors find it most practical to read about the next point of interest before coming to it, often having someone other than the driver read this aloud while traveling.

In the left hand margin of the text is a series of numbers which correspond to numbered markers along the road marking the point of interest. Explanations follow each number in the text. The fume cloud on the roadside markers points in the direction of the feature. If you are entering via Manzanita Lake Entrance Station, start from the last of this book, and read the numbers in descending sequence.

The mileage shown in parentheses in the center line of the pages indicates the distance between successive points driving either way.

ALWAYS PARK OFF THE PAVEMENT
AN AUTO ACCIDENT WILL SPOIL YOUR TRIP



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National Park Service
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INTRODUCTION

Lassen Volcanic National Park was established by act of Congress in 1916 from lands of the Lassen National Forest and the small Lassen Peak and Cinder Cone National Monuments formerly under the Forest Service, U. S. Department of Agriculture. The park area now measures approximately 10 by 17 miles, embracing about 165 square miles.

The Lassen Park Road is a link in State Route No. 89. It runs through Lassen Volcanic National Park, half circling the east side of Lassen Peak. This road was built by the National Park Service of the U. S. Department of the Interior to make accessible by car some of the finest scenery in the park. The road is not designed for high speed. The slower you drive the more you will see. Like you, others are sight seeing, so drive carefully and observe the warning signs at all times. Remember: an accident may ruin your vacation, your car, and even your life. Maximum speed is 45 miles per hour, but in many places this is excessive.

While using this guide you will want to stop many times. Be sure to park OFF THE PAVEMENT to prevent accidents. In the spring when heavy snowdrifts line the road, it will not be practical to stop in certain areas, nor will it be possible to see some of the things described in this publication.

Other books dealing in greater detail with the interpretation of the natural and human history of the area may be purchased in the Visitor Center at Manzanita Lake.

A MAP OF THE PARK WILL BE FOUND ON PAGES [18](#) and 19



Look for numbered stakes along the road.

1 THE RAKER MEMORIAL was sponsored by the California State Chamber of Commerce in 1931 in honor of Congressman John E. Raker who was Representative of the State for 15 years. It was he who introduced the successful bill to Congress recommending the establishment of Lassen Volcanic National Park. The bill was signed by Pres. Wilson on August 9, 1916. John Raker died in 1926. Appropriately, this is also the boundary between the National Park and Lassen National Forest. Although the National Park Service and the National Forest Service differ in conservation policies, they cooperate very closely on mutual matters, such as forest fire and forest pest control.



(0.5 mile)

2 THE BROKEOFF MOUNTAIN TRAIL starts here on the uphill side of the road. It is a 3.5 mile hike to the summit at 9,235 feet. The elevation here at the road is 6,640 feet.

The common shrub-like trees along the first part of the trail are the mountain alder and willow. (see sketch). In the fall the willow leaves turn bright yellow. Both species lose their leaves in winter.



MT. ALDER



WILLOW

(0.4 mile)

3 THE NEW SOUTHWEST ENTRANCE STATION was 2 completed in the summer of 1966. Entry permits, which are required during the summer season, are sold here.

(0.3 mile)

4 THE VISITOR USE BUILDING was also completed in the summer of 1966. It is designed primarily for winter use with the ski slopes nearby. Lunches, souvenirs and information may be obtained here in the summer season also.

(0.8 mile)

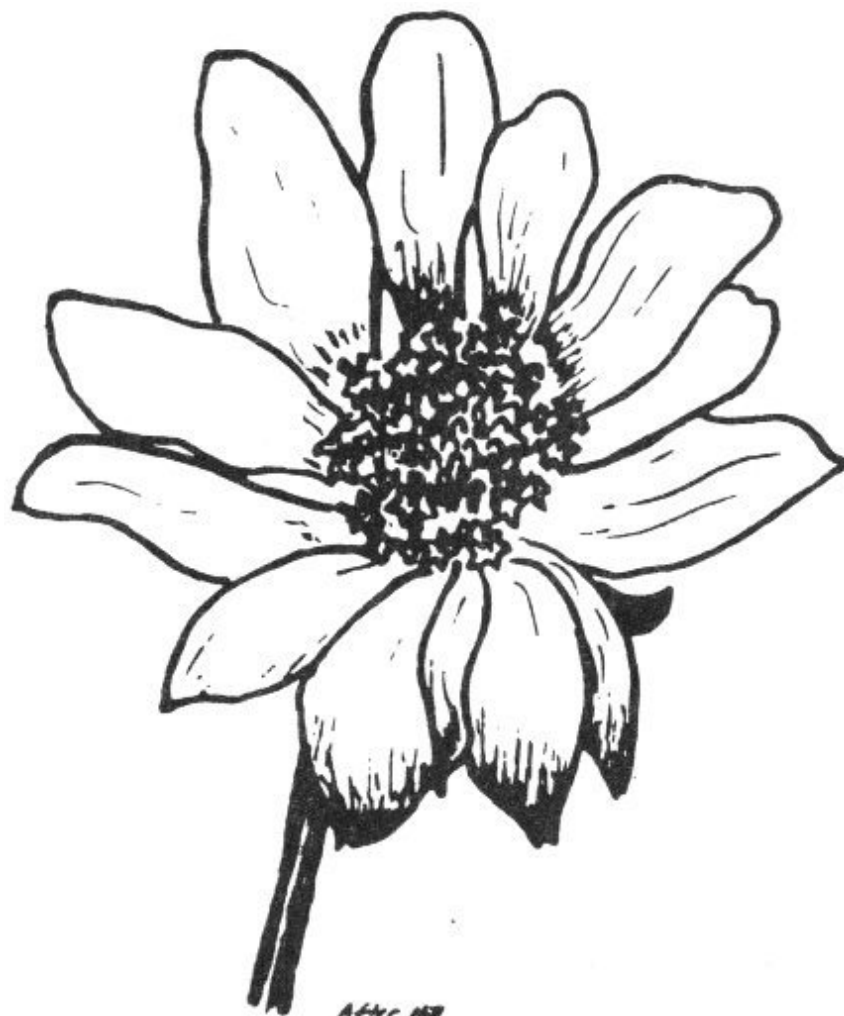
5 SULPHUR WORKS is the most easily accessible hydrothermal area in the Park. It is probably part of the central vent system of ancient Mt. Tehama which is described at No. 7. A short paved trail leads to the sputtering hot springs, steaming fumeroles and hot bubbling mud pots. For your own safety be sure to stay on the trails in any of the thermal areas. Slippery clay and thin crusty coverings could lead to a dunking in scalding hot water and mud. Most water in the thermal areas of the Park contain sulphurous or sulphuric acid, so avoid getting it on your clothes. The odor is mainly that of hydrogen sulphide. Much of the white clay is tinted yellow, tan or pink by mineral impurities, chiefly iron oxides.

The name Sulphur Works was first used in 1865 when efforts were made by T. M. Boarman and Dr. M. Supan to develop the sulphur and clay potentialities of the area. This land was filed upon originally as a mining claim but was actively used only for the tourist trade beginning about 1940. Some of the non-acidic thermal water was used for hot baths. The property was acquired by the Federal Government in 1951 and added to the National Park to be used as a natural thermal area exhibit.

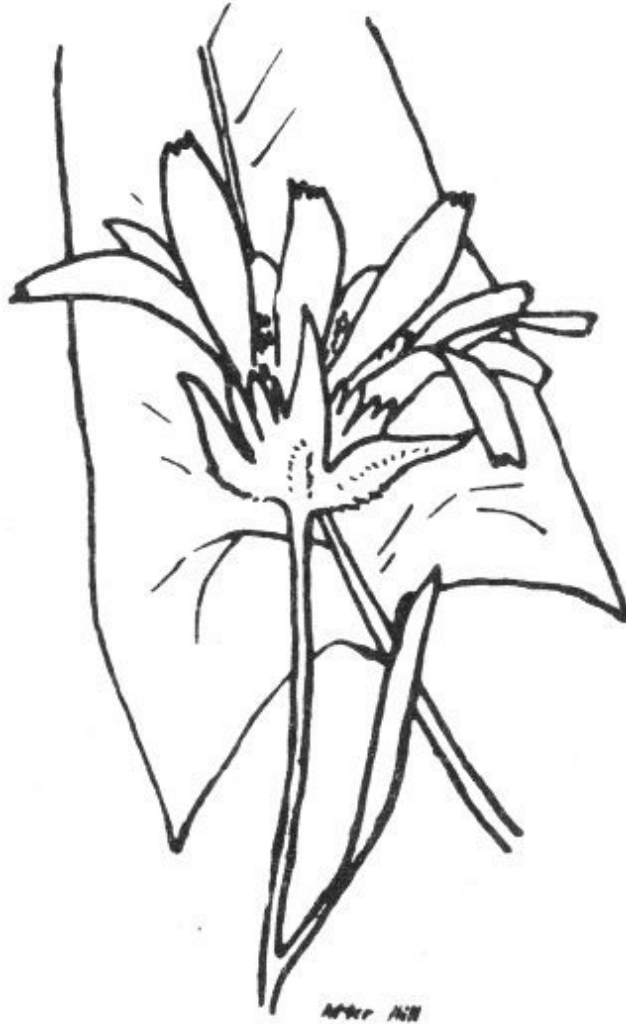
(0.3 mile)

In early summer the flowers along the road for the next mile or two, are spectacular. Yellow balsam root (with arrowhead-shaped leaves), mules ears (with similar blossoms), and blue stickseed, which looks like forgetmenot, are the most showy. In late summer tall stalks of false hellebore and yellow senecio are common.

(0.1 mile)

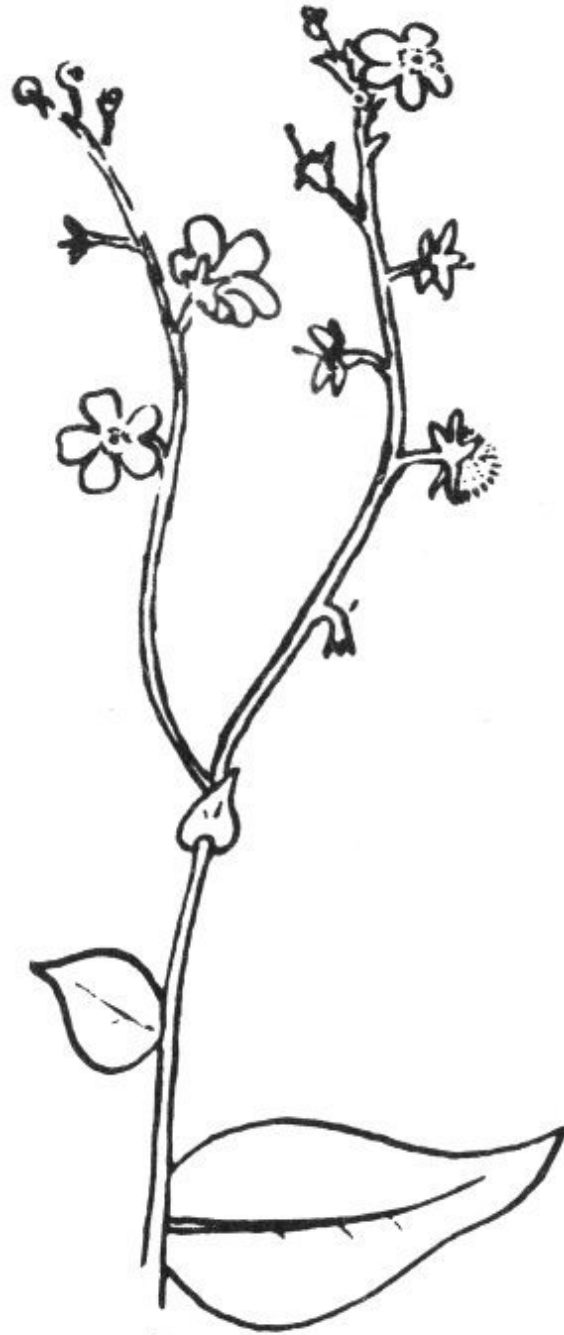


MULES EARS (yellow)



After Hill

BALSAM ROOT (yellow)



After Hill

BLUE STICKSEED

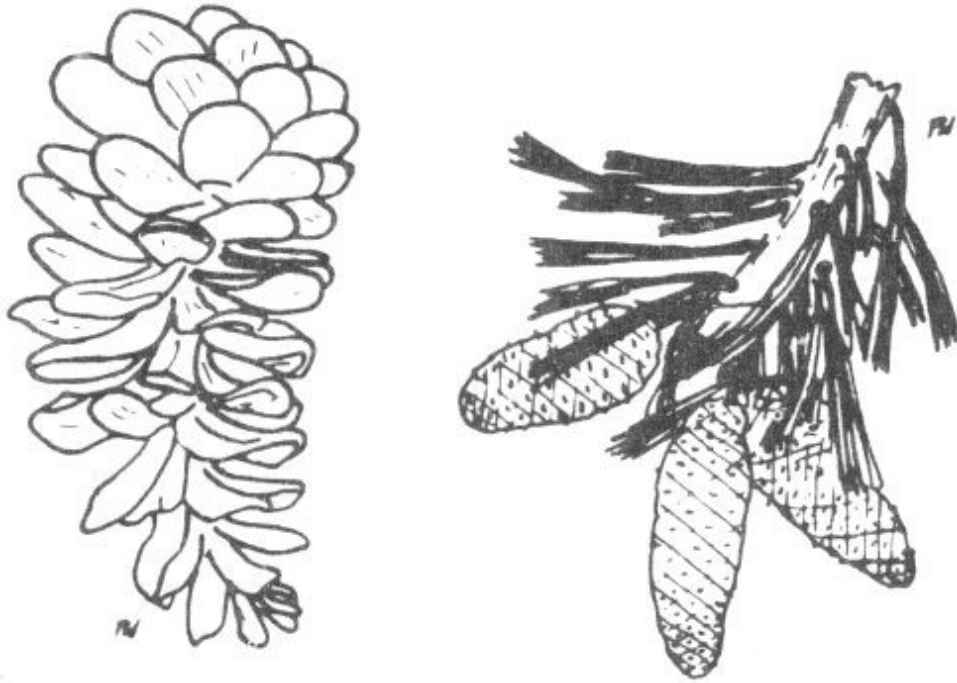


SENECIO (yellow)

6 ELEVATION 7,000 FEET. Just ahead is a magnificent specimen 4 of western white pine. Being one of the white pine group, it has five needles in a bundle. Its cone is about $\frac{1}{3}$ the size of its near relative of lower

(Transition Life Zone) elevations, the sugar pine. Its limbs grow far down on the trunk because they get plenty of light for the manufacture of food. Shading in dense forests causes a sloughing off of lower limbs, resulting in natural pruning.

(0.4 mile)



WESTERN WHITE PINE (five needles)
Mature Cone, 6 in. Young Cones and Foliage

7 BROKEOFF MOUNTAIN, the second highest peak in the park. The volcanic lava and ash layers are abruptly broken off on the north side, hence the name "Brokeoff". This mountain is a flank remnant of ancestral Mount Tehama which once towered more than 1,000 feet higher than Lassen Peak. Tehama was a composite, or strato-type volcano like Mt. Shasta. It was destroyed by a series of cracks, called faults, which cut Tehama into huge blocks. These sunk, causing collapse of the great mountain. Mt. Conard to the southeast, and the ridge between Brokeoff and Lassen Peak, are also remnants of the Tehama rim, all dipping away from the center of the old volcano.

On the southeast horizon are the Sierra Nevada, with Childs Meadow closer in. To the north is the ragged top of Diamond Peak with a natural window through the rocks on the upper right near the skyline.

(0.6 mile)

8 DIAMOND POINT, on the southeast flank of Diamond Peak. 5

This rock is volcanic agglomerate, a mass of volcanic ejecta which became stuck together by small amounts of still molten lava. Nearby is one of the volcanic vents of ancestral Mt. Tehama. The material on top of Diamond Peak is explosive debris slightly consolidated by the cementing action of volcanic ash and is called tuff. It has weathered into very rough and rugged forms.

There is an excellent view of Mt. Conard.

Mill Creek Canyon below has been glaciated; Childs Meadow is visible at the mouth of the canyon.

(0.4 mile)

9 LASSEN PEAK SIGN. This is one of the best views of the south side of Lassen Peak. To the left of it is sharp Eagle Peak and rounded Ski Heil. To the right, across the canyon at eye level, the red coloring is due to iron oxides, and the yellow-green is due to millions of lichen plants on the rocks. Across the roadway at the roadbank is cool drinking and car radiator water. Watch for autos when crossing the highway!

(0.1 mile)

10 THE SLOPING LAYERS OF LAVA AND ASH across the canyon to the east were part of ancestral Mt. Tehama. Just to the right are highly colored rocks that were once grey andesite lavas, but they have been chemically altered by steam and sulphurous acid from below. The white areas are largely clay with some non-precious opal. The tans, red, and purple

are stains of iron oxides. Volcanic fumes reach the surface abundantly in this area, rising along fault cracks. Deep in the canyon below lies picturesque little Ink Lake.

(0.2 mile)

11 THE HUGE BOULDERS on the west (uphill) side of the road are chunks of cemented volcanic fragments which have rolled down from the top of Diamond Peak.

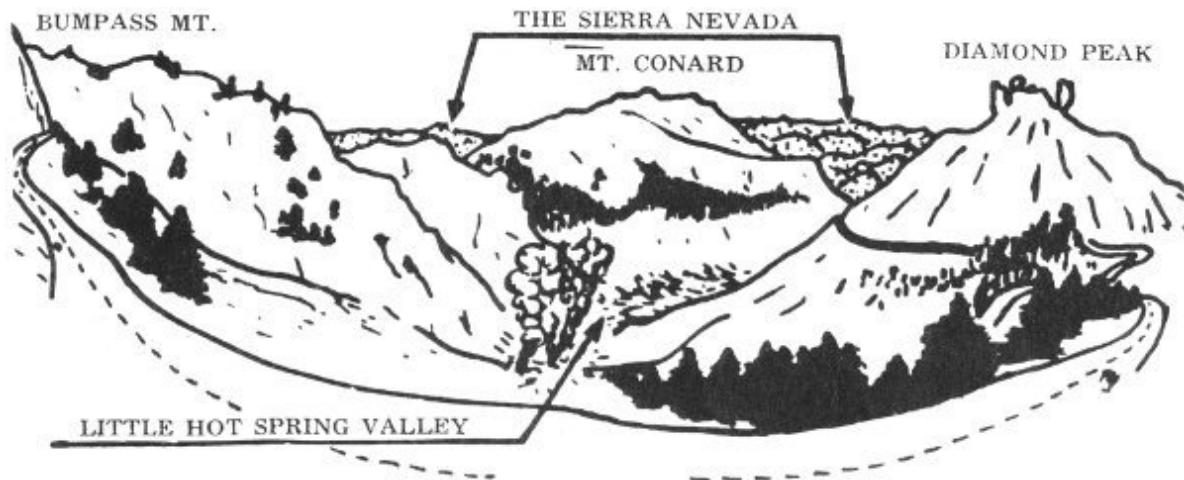
Lassen Peak is visible to the north (up canyon) from this vicinity, 6 picturesquely framed by roadside trees.

(1.4 miles)

12 Rugged DIAMOND PEAK below, nearly encircled by Lassen Park Road, was named for the occurrence of occasional small, double-ended quartz crystals and diamond-shaped calcite crystals. These were deposited by hot volcanic waters that seeped through its already hardened lavas after volcanic eruptions had ceased. The mountain is the tough lava feeder, or core, of prehistoric Mount Tehama. Diamond Peak has resisted the sinking and weathering which reduced its surroundings.

(0.7 mile)

13 HEAD OF LITTLE HOT SPRINGS VALLEY. The “U”-shape cross section of the canyon indicates glacial widening in contrast to the “V”-shape found in non-glaciated stream cut canyons. There is another hot spring area above, near the base of Pilot Pinnacle.



Landscape

THE SIERRA NEVADA
 BUMPASS MT.
 MT. CONARD
 DIAMOND PEAK
 LITTLE HOT SPRING VALLEY

(0.3 mile)

14 EMERALD POINT: A few yards away is the best view of the southwestern portion of the park. Left to right is Mt. Conard, Little Hot Springs Valley, Diamond Peak, Brokeoff Mtn., Mt. Diller, Pilot Pinnacle, Ski Heil, Eagle Peak, and Lassen Peak.

In his book “The Mountains of California”, John Muir wrote of the lovely mountain hemlocks. “Some of the finest groves I have yet found are on 7 the south slopes of Lassen’s Butte.”

(0.2 mile)

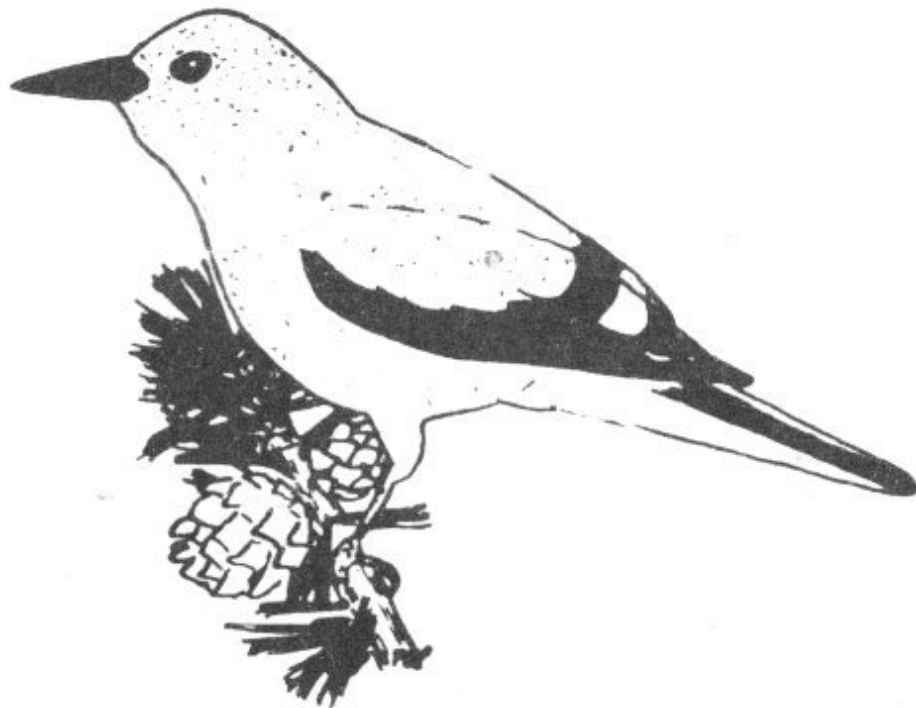
15 EMERALD LAKE, of glacial origin, was named for its beautiful green color. Fishing is not allowed in this lake, and since it is planted with fish, large Rainbow Trout can be seen close to shore. Please do not spoil the

beauty of this spot by littering the lake or the shore with tin cans and other refuse.

This point is just over 8,000 feet elevation, in the Hudsonian Life Zone. Here the mountain hemlock trees, with their graceful nodding tops, and the noisy grey, black, and white jay known as the Clark's nutcracker, are most conspicuous and typical forms of life.

The smooth slopes on Ski Heil Peak behind Emerald Lake are unexcelled for skiing.

(0.3 mile)



CLARKS NUTCRACKER (grey with black and white)



MT. HEMLOCK (immature purple cones)

16 GLACIAL ERRATIC. This great isolated lava boulder, perched on the outside edge of the road just south of the Bumpass Hell parking area, was carried by a glacier from the southeast base of Lassen Peak and deposited here when the ice river melted. The rock on which it lies has a scratched, grooved, and highly polished surface. This is the work of rocks frozen into the base of the thick glacier, which moved over this area and into the valley below during the Ice Age. There are few places where such evidence of volcanic heat and glacial ice is found. Please park at [No. 17](#) if you wish to walk to the Glacial Erratic.

8

(0.1 mile)



Landscape

17 BUMPASS HELL PARKING AREA. The Bumpass Hell self-guiding nature trail, with explanatory leaflets provided enroute, is a 1.1 mile walk to the edge of the hot spring basin. This is Lassen's most spectacular and diversified hydrothermal area. The walk through it involves another mile. In midsummer wildflowers are excellent along the trail: bog kalmia and Brewer mountainheath (low pink flowering shrubs), coast erysimum (orange), Newberry penstemon (red), and silverleaf lupine are the most conspicuous. Pinemat manzanita (white), rabbitbrush (yellow), and chinquapin (cream-colored) are dwarf shrubs which are also common.



BOG KALMIA (pink)



COAST ERYSIMUM (orange)



BREWER MOUNTAINHEATH (deep rose)

9



SATIN LUPINE (blue)



NEWBERRY PENSTEMON (dark red)

You can hike through Bumpass Hell and continue via Cold Boiling Lake to Kings Creek Campground. It is an easy walk of 4 miles, most of which is downhill.

(0.2 mile)

18 LAKE HELEN is a deep, blue, glacial lake at 8,164 feet elevation. It is frozen over for 7 or 8 months a year, and is very cold even in summer being 39 degrees at depth. Lake Helen is exceptional among Lassen's lakes in that fish plantings here have been unsuccessful ... perhaps because of a lack of native food.

This body of water was named by Major Pierson B. Reading for Mrs. Helen Tanner Brodt who ascended Lassen Peak with him in 1864, the first woman known to have made the climb.

Lassen Peak is across the lake to the northwest. The cliffs represent portions of the original plug of stiff, pasty dacite lava which was forced up rapidly as

a unit through the crust of the earth. Along the right shoulder of Lassen the trail zig-zags up the mountain.

On the side of the road away from the lake is a large andesite lava outcrop of vertical plates, or slabs. Known as jointing, this has been caused by strains set up in the cooling lava mass after it hardened. This helps to wear away mountains because water seeps into these cracks and wedges them apart when it freezes and expands.

(0.2 mile)

19 LAKE HELEN PICNIC AREA. In 1933 a bronze plaque was 10 placed here by Ethel Brodt Wilson and her children through the sponsorship of the Shasta Historical Society. It reads: “Lake Helen, elevation 8,164 feet. Named for Helen Tanner Brodt by Major Pierson B. Reading in honor of her being the first white woman to see the lake and to make the ascent of Lassen Peak August 28, 1864....”

Just south is the snow-measuring course, identified by yellow and red markers in the hemlock trees and a tall iron pipe.

(0.2 mile)

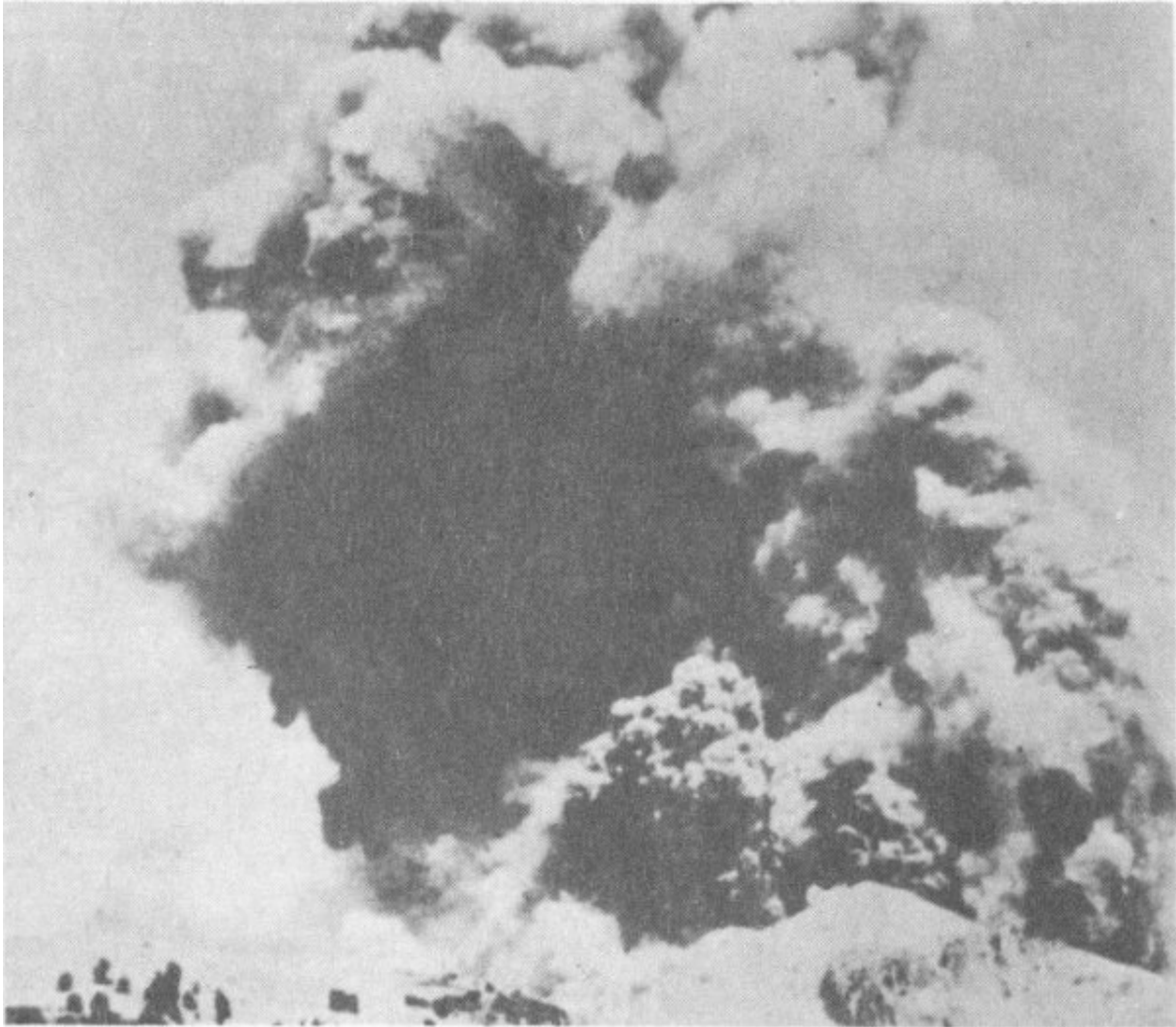
20 The Park Service WINTER CONTROL CABIN is a hundred yards east (uphill) of the road. It is used by Park Rangers on ski patrols and on the monthly winter snow-measuring trips. The latter are conducted in cooperation with the State of California Snow Survey. The depth of snow pack and the amount of water it contains determines the schedule of commercial water-use activities in the valley for the following summer. Snow packs of 20 feet representing a snowfall of over 50 feet, are common in this area.

(0.4 mile)

21 Scenic view of LAKE HELEN and the western half of the REMNANTS OF COLLAPSED MT. TEHAMA. This ancestral mountain, a composite or strato-type volcano, was destroyed during the Ice Age in much the same manner as Mt. Mazama, which collapsed to form 2,000 foot deep Crater Lake. Brokeoff Mt., Mt. Diller, and Pilot Pinnacle to the south and west are remnants of Mt. Tehama.

In the roadcut to the east of rounded markings are “inclusions” of an early stage of hardening of the crust of this dacite lava which was later shattered, and engulfed, and partly remelted when the molten rock from below forced its way upward.

(0.3 mile)



**“Vulcan’s Face” appeared in a 1914 eruption cloud over Lassen Peak.
—photo by a U. S. Forest Service Ranger**



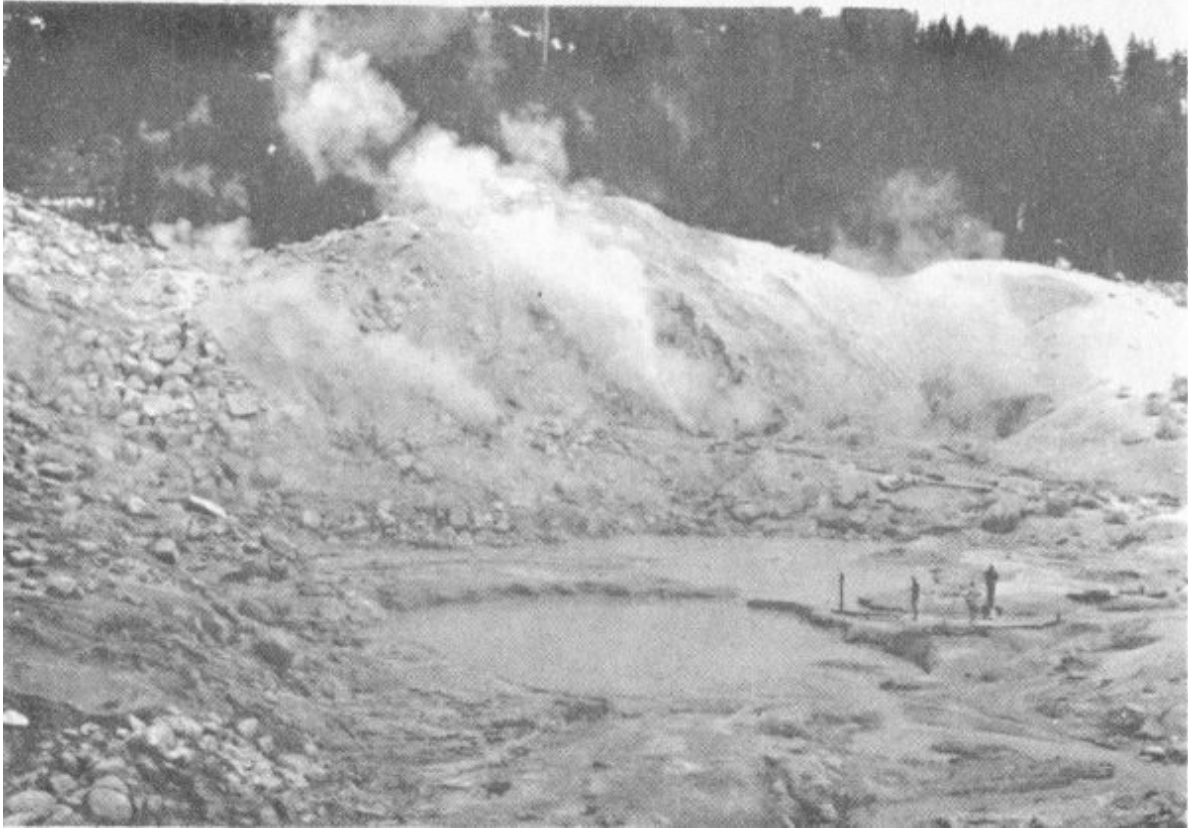
The Devastated Area, [No. 44](#) in the Road Guide, is slowly recovering after eruptions in 1915 flattened trees on the northeast slope of Lassen Peak.

—NPS photo by R. C. Milne



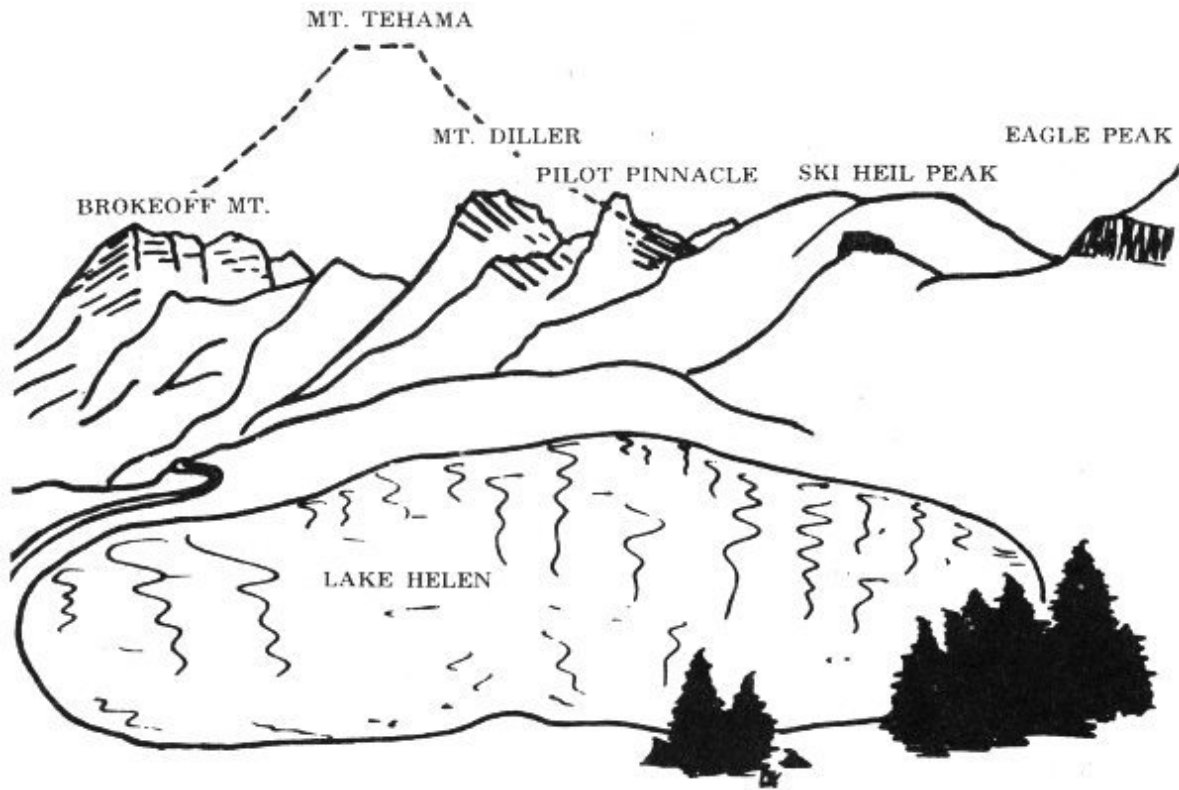
Bumpass Hell, the most spectacular thermal area in the Park, is 1,400 feet long and 500 feet wide.

—NPS photo by R. L. Nelson



A self-guiding trail, starting at [No. 17](#) in the Road Guide, leads to steaming fumeroles, hot springs and boiling mudpots.

—NPS photo by R. L. Nelson



Landscape

BROKEOFF MT.
MT. TEHAMA
MT. DILLER
PILOT PINNACLE
SKI HEIL PEAK
EAGLE PEAK
LAKE HELEN

22 LASSEN PEAK TRAIL SIGN. On the southeast side of the road is the Peak Parking Area, and on the northwest side starts the trail up Lassen Peak. The Peak Trail is a well-graded climb of about 2,000 feet: from 8,500 feet to 10,457 feet in 2½ miles of hiking. It takes most persons about 4 hours for the round trip, though some take an hour more, and a few an hour less. For a pleasant hike:

1. Wear low heeled, sturdy shoes.

2. Hike at a moderate pace, taking short, frequent rests and enjoy the ever-changing view.
3. Take a lunch along.
4. You can eat snow for water, but do so slowly, otherwise it may make you ill.
5. Dark glasses, sunburn lotion, and a hat are advisable.
6. Do not start later than 2 hours before sunset, or when storms or fog threaten.
7. **DO NOT TAKE SHORT-CUTS ACROSS THE TRAIL.** It is against Park Regulations and is inadvisable because:
 - a. It is dangerous: sprained or broken ankles and arms may result.
 - b. Unavoidable dislodging of rocks is dangerous to hikers below.
 - c. It costs money to rebuild the trail which short-cutting ruins.

On the rise to the left of the trail, the parallel markings in the rock is known as flow banding and takes place when the lava is still moving but cooling rapidly. Directly above this banding, on top of the rock formation, is glacial polish, which indicates that the lava cooled long before the Ice Age. 12

In climbing the Peak, the trail leaves the Hudsonian Life Zone with mountain hemlocks and white-bark pine, and at timber line it enters the Arctic-Alpine Life Zone.

From near the register box atop Lassen Peak, Mt. Shasta is to the northwest, 75 miles away, looming up 14,161 feet. To the north, are Chaos Crags, the Devastated Area, Prospect Peak, Cinder Cone, and Butte Lake. Mt. Harkness, Warner Valley, and the mountains of Nevada are to the east, with Dyer Peak and Lake Almanor in the distance and Kings Creek Meadow nearby. To the southeast are the High Sierra including Pyramid Peak in the vicinity of Lake Tahoe.

To the west is the rough black dacite lava of 1915 which filled and obliterated Lassen's 1914 Crater. This lava also spilled through the northeast notch, causing the Great Mudflow of May 19, 1915. Just beyond the 1915 lava is the 1915-1916 crater formed by the great explosive eruption of May 22, 1915. It now contains a tiny lake, sapphire blue when not frozen over. The notch on Lassen's western skyline, just beyond, is the 1917 crater. There

are active steam vents in the west and north portions of this basin, but the steam does not always condense well enough to be seen easily. There are yellow deposits of sulphur on the north wall of 1915-1916 crater.

On the west side of the crater basin, the 1915-1916 and the 1917 craters can be seen, and also the Brokeoff-Lassen Ridge to the south, glaciated Blue Lake Canyon to the southwest, Loomis Peak and the Sacramento Valley to the west, with the Coast Range Mountains beyond. The highest Coast Ranges to the northwest are the Trinity Alps. Nearby is Manzanita Lake area with Chaos Jumbles and Chaos Crag to the right (north). 13

Lassen Peak was named after Peter Lassen, a pioneer of Danish birth who is reported to have used the mountain as a landmark. Lassen blazed a round-about emigrant trail from Black Rock (Nevada), east and south of the present park area, to his Deer Creek Rancho “Bosquejo” in the Sacramento Valley.

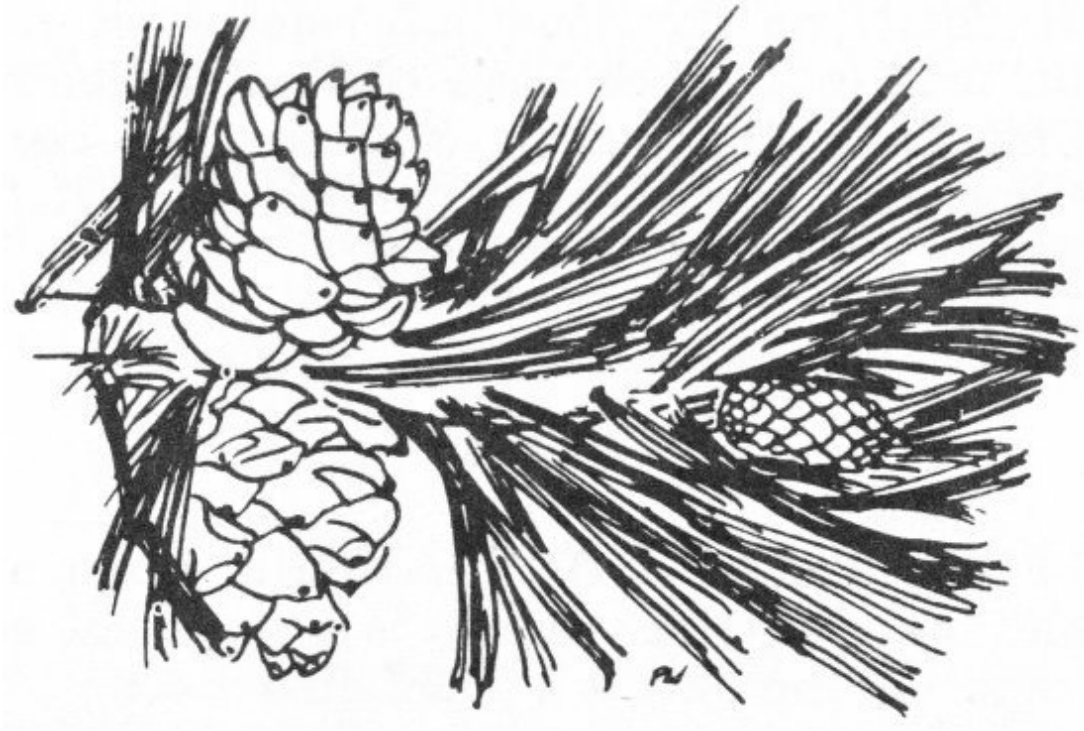
(0.2 mile)

23 SUMMIT SIGN, HIGHEST POINT on the road: 8,512 feet above sea level.

For about a mile either way along the road in spring and early summer cross country skiing is normally good. Snow generally closes the road for the winter in November and plows cut through the winter pack anywhere from the end of May to early July.

(0.4 mile)

24 In addition to the mountain hemlocks, here also are the gnarled, multiple-trunked WHITE BARK PINES. This is one of the few places in California where this fascinating and photogenic tree can be reached by road. The snow banks seen on the rocky slopes across the ravine frequently last through the whole summer.



WHITE BARK PINE (cones 1½ inch)

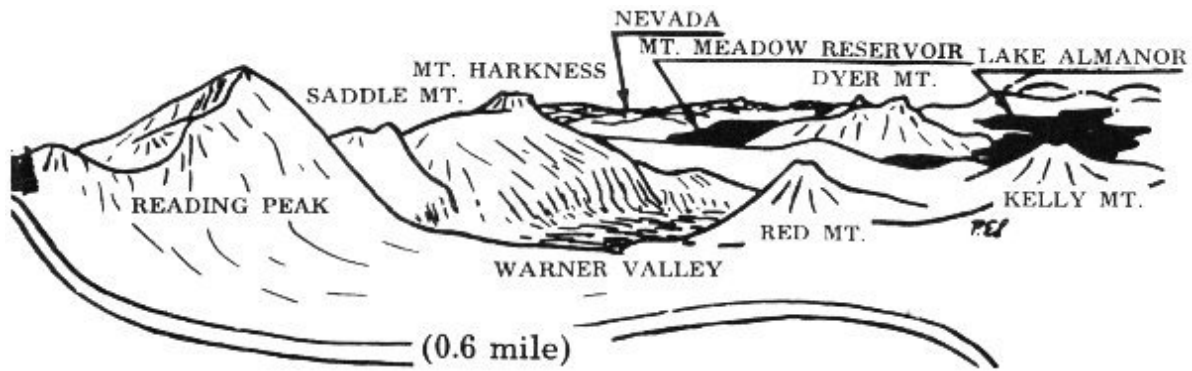
(0.3 mile)

25 LAKE ALMANOR AND MT. HARKNESS VIEW SIGN.

14

Warner Valley has been modified by a glacier 1,000 feet thick. Lake Almanor and Mountain Meadow Reservoir, both outside the park, are artificial lakes, part of the Caribou Power System of the Pacific Gas and Electric Company. Peter Lassen's emigrant trail from Black Rock, Nevada, ran from north to south through the Lake Almanor basin (Big Meadow) before turning west again.

Highway No. 36 runs between the Cascade Range on the north and the Sierra Nevada on the south.



Landscape

NEVADA
 MT. MEADOW RESERVOIR
 LAKE ALMANOR
 MT. HARKNESS
 DYER MT.
 SADDLE MT.
 READING PEAK
 KELLY MT.
 RED MT.
 WARNER VALLEY

(0.6 mile)

26 FISSURES may be seen in rocks 100 yards from the road and also near the road between here and stake 27. These cracks, sometimes quite large, are typical of weak zones in volcanic areas.

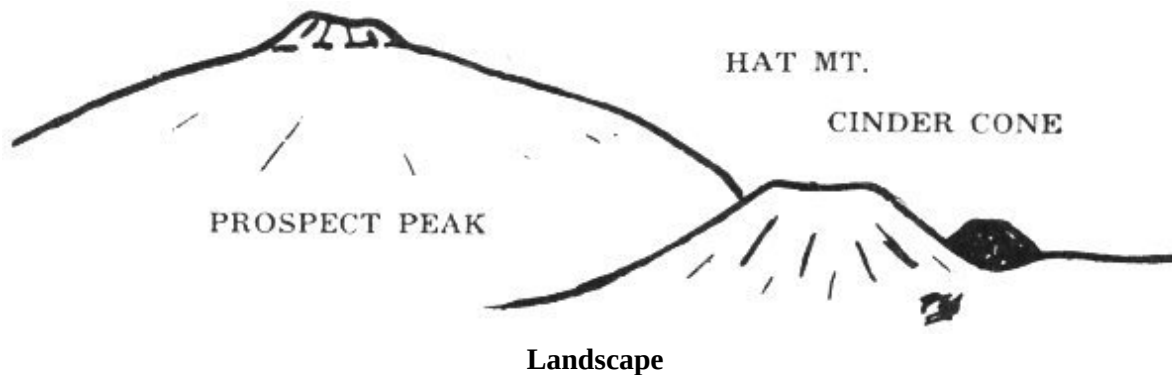
(0.5 mile)

27 TRAIL SIGN on the down-hill side of the road indicates the start of the trail to Terrace Lake (0.5 mile); Shadow Lake (0.8 mile); and Cliff Lake (1.5 miles). You can also reach the park road again at Summit Lake about (4.0 miles) or at Hat Lake about (3.5 miles).

(0.1 mile)

28 8,000-FOOT ELEVATION MARKER. From here Prospect Peak is visible. Like Mt. Harkness, it is a small but good example of shield volcano. On top of it is a small cinder cone which destroys the low shield-shape cross-section (indicated by the dotted line) characteristic of that type of volcano. At the right base of Prospect Peak is Hat Mountain, and behind it, at its right base, remarkable Cinder Cone in the northeast corner of the park.

15



HAT MT.
CINDER CONE
PROSPECT PEAK

(1.0 mile)

29 A fine view of UPPER KINGS CREEK MEADOW. This may well have been a lake once, filled in by glacial and stream-borne materials. The level meadow floor caused Kings Creek to move slowly and to wander about “aimlessly” in a pattern called a “meander” by geologists. Where it leaves the flat meadow the stream takes on direction and speed, thus resuming a relatively straight, swift course. As the meadow builds up higher, becoming drier, the encroaching forest will eventually engulf it.

(1.3 miles)

30 KINGS CREEK CAMPGROUND and Picnic Area. This is the Canadian Life Zone, although a few “Hudsonian” mountain hemlocks are in the cooler portions. The 7,400 foot elevation makes this a most delightful, primitive type of campsite during the warm summer months.

The trail goes south from the campground to Cold Boiling Lake (0.8 mile) and to Crumbaugh Lake (1.5 miles).

Kings Creek, originating from the springs in these meadows, is a part of the Feather River system, draining all of the area between Lassen and the Truckee River Divide, the largest single tributary drainage area in California.

(0.5 mile)

31 READING PEAK rises over 1,300 feet to an 8,701-foot elevation. Like Lassen Peak, across the meadow to the left, this is a plug volcano. It honors Major Pierson B. Reading (pronounced Redding), Northern California pioneer. He was General Fremont’s paymaster, and was involved in the Bear Flag revolt. Reading also discovered gold on Clear Creek in March 1848, and was the first permanent settler of Shasta County. In 16 1864 he led the third recorded ascent up Lassen Peak. The same year he filed a mining claim jointly with K. V. Bumpass on the Bumpass Hell area.

Deer are generally seen here in the meadow, especially near sunset during late summer and early fall.

(0.6 mile)

32 KINGS CREEK. On the west side of the road is a small picnic area. On the downstream side of the road (left of the creek) is a trail to Warner Valley. Kings Creek Falls is 1.3 miles, and Drakesbad is 4.0 miles. You can hike uphill to the base of Lassen Peak.

In both Lower and Upper Kings Creek Meadow, fawnlilys, red heather, corn lily, and white gentian make a fine floral succession from early spring to autumn. Lupine and pussy paws are also common in the drier locations.

(0.2 mile)



FAWNLILY (yellow and pinkish)



FALSEHELLEBORE (whitish)

33 An EMERGENCY TELEPHONE is just below the park road for reporting accidents and fires.

17

(1.0 mile)

34 LAKE ALMANOR, WARNER VALLEY, and what might be called the friendly wilderness of the eastern portion of the park. Trails reach this country from Summit Lake and from dead-end secondary roads at Butte Lake, Horseshoe Lake, and Warner Valley. On Mt. Harkness is a National Park Service fire lookout station. Help keep the forests green by being very careful with matches, cigarettes, and the like.

(0.8 mile)

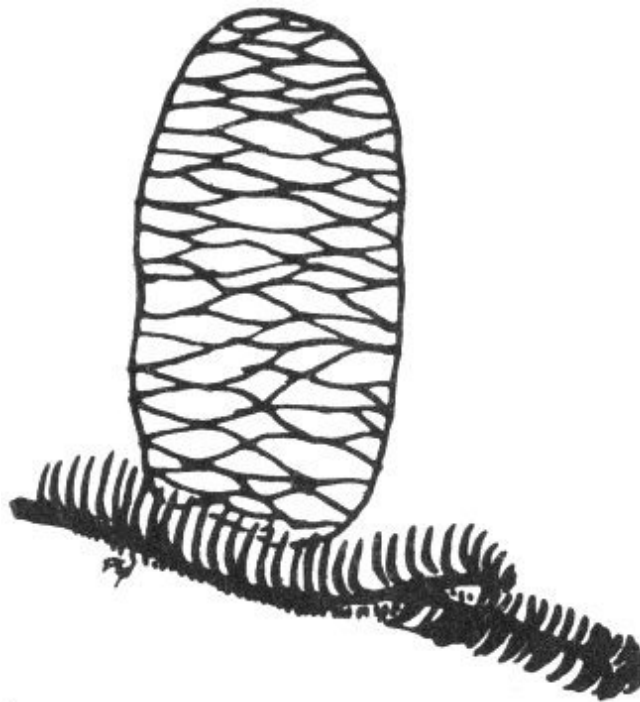
35 In this vicinity once stood a fine stand of RED FIR. On October 11, 1962, winds of gale force, sweeping up through the valleys from the southeast, blew down and broke off many large trees. This is typical of the story of violence and peace of the Lassen area. In a few years the young growth will hide most of the devastation.

The short, blue-green, single needles of the red fir grow in a brush-like manner. Very symmetrical, the trees are called “Silver Tips” in the commercial Christmas tree market.

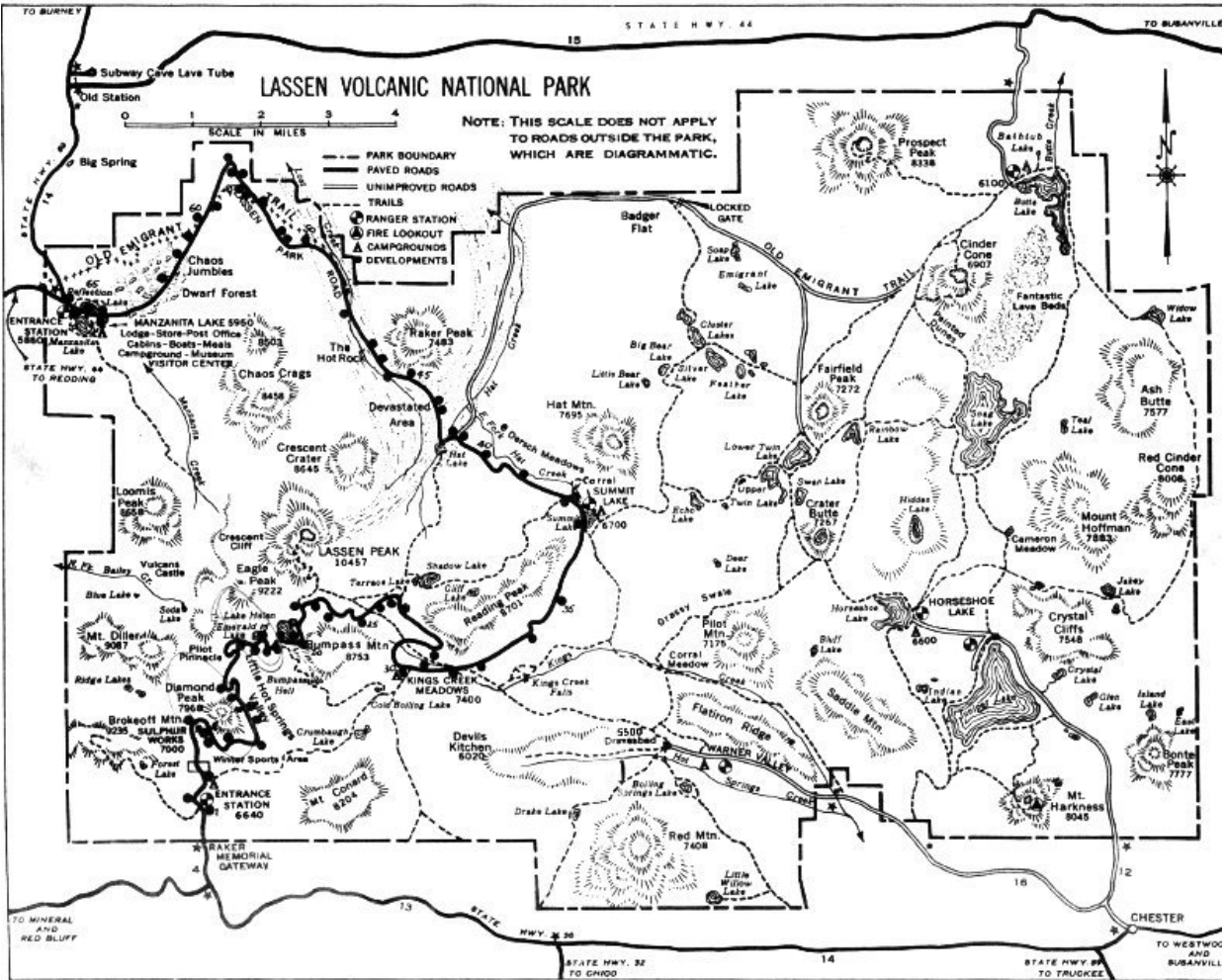
The trunks curve at their base due to the heavy snowfall which bends the saplings downhill, flat to the ground. In the spring the young trees curve up, but the bend remains, and persists throughout the life of the tree. Curved trunks can also be found in other species of trees, such as mountain hemlock.



CURVED TREE TRUNK



RED FIR



LASSEN VOLCANIC NATIONAL PARK

[High-resolution Map](#)

20

(1.4 miles)

36 SUMMIT LAKE lies on the east side of the road with entrance roads to the North and South Campgrounds, 0.1 mile apart. The lake is so named because it lies at the very summit of a divide between the Feather River drainage to the south and the Pit River drainage to the north.

The elevation of the lake is 6,700 feet. It is good for swimming, and its shores offer fine spots for picnicking or camping. Fishing is generally only fair.

This area is dominated by red firs, with lodgepole pine and western white pines. Also characteristic is the pinemat manzanita which forms a carpet on the forest floor, bearing numerous small white blossoms. Cream-colored marshmarigolds, and later white false hellebore, are the most showy of numerous flowers to be seen in moist areas about the lake.

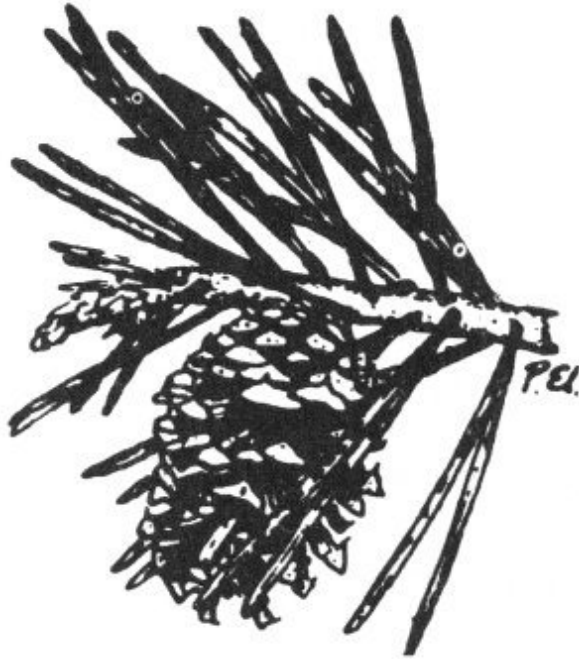
Hat Mountain is the nearby symmetrical, flat-topped cinder cone, rising 1,195 feet above the level of Summit Lake, to the northeast of it.

A trail circles the lake and on the far side of it, heads east to Echo Lake, Twin Lakes, Cinder Cone, and Butte Lake; or to Bear and Cluster Lakes; or to Kings Creek Falls, Warner Valley, and Drakesbad.

(0.4 mile)



MARSHMARIGOLD (cream-colored)



LODGEPOLE PINE (cone 1¼ inches)

37 SUMMIT LAKE RANGER STATION. An emergency 21 telephone is located there.

(0.1 mile)

38 DRIVE SLOWLY: TRAIL CROSSING FOR HORSES. The side road on the east leads to the public horse corral. Reservations for riding and pack horses may be made during the summer season here or at the Manzanita Lake Lodge.

The trail on the west goes to Cliff Lake (2.3 miles), Shadow Lake (3.2 miles), Terrace Lake (3.5 miles) and on to the base of Lassen Peak.

(1.2 miles)

39 DERSCH MEADOWS is a two mile long series of meadows extending northwest from Summit Lake along the park road. It is drained by

the East Fork of Hat Creek and Dersch Creek. A large variety of fine spring and summer wildflowers including alpine shooting stars, columbia monkshood and rare white orchids, milkwhite habenaria, can be found close to the road. Deer are abundant here during the summer and fall seasons.



COLUMBIA MONKSHOOD (blue, rarely white)



MILKWHITE HABENARIA (white)



ALPINE SHOOTING STAR (strong pink)

This is CANADIAN LIFE ZONE country. In addition to red fir, the 2-needle lodgepole pine, and western white pine are abundant. The shrub-like trees in the meadows are the moisture-loving willows. The rust-brown branches on the red fir have been caused by a small mistletoe which, like most mistletoes, cripples but rarely kills the trees.

22

(0.3 mile)

40 A large ANT HILL at the base of a small lodgepole pine about 15 feet from the road on the uphill side is about a 3-foot high nest built of twigs and dry pine needles by red ants about ¼ inch long. These ants have a very well developed social system comparable to that of bees. PLEASE DO NOT POKE OR OTHERWISE DISTURB THE ANTHILL.

(0.7 mile)

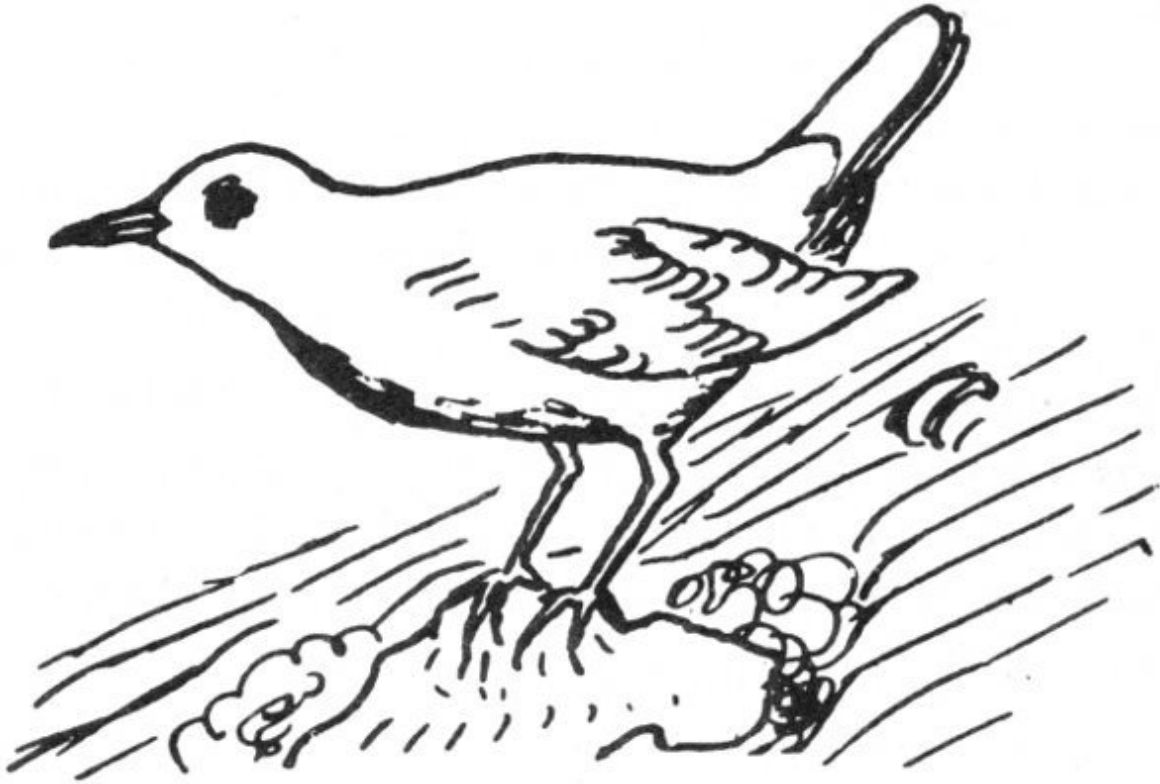
41 On the north (outside) edge of the park road is a truck trail which follows HAT CREEK north, and then eastward to a locked gate at Badger Flat, 6 miles distant. From Badger Flat, trails lead to Cinder Cone (5 miles), Butte Lake (7 miles), Cluster Lakes (2 miles), and many other points.

This is the southeast boundary of the DEVASTATED AREA. The down logs all point away from Lassen Peak, which dominates the scene and which wrought the devastation. Some trees have been uprooted; others have been snapped off like match sticks.

(0.2 mile)

42 HAT LAKE PARKING AREA. Nearby is Hat Lake, elevation 6,450 feet. It was formed by the mudflow of May 19, 1915, when it blocked Hat Creek. All lakes are very temporary features, and this one is a good example, as it is rapidly being filled by the building of a delta at the south end of the lake and by the accumulation of organic debris such as logs, etc. This will become a meadow in the near future. Most meadows have been formed in a similar manner. A cream-colored aquatic buttercup blooms conspicuously on the lake in the summer and early fall.

This is a good place to observe birds. The dark, grey-brown water ouzel, or dipper, can usually be seen or heard near the outlet where Hat Creek is crossed by the Park Road.



DIPPER or WATER OUZEL (grey-brown)

Just across the road from the parking area, the trail to the south goes to Paradise Meadow: (1.5 miles). An excellent variety of wildflowers bloom along the trail in mid-summer. Terrace Lake is 2.9 miles from here, and the road is again reached at an elevation of nearly 8,000 feet, 0.5 miles beyond this.

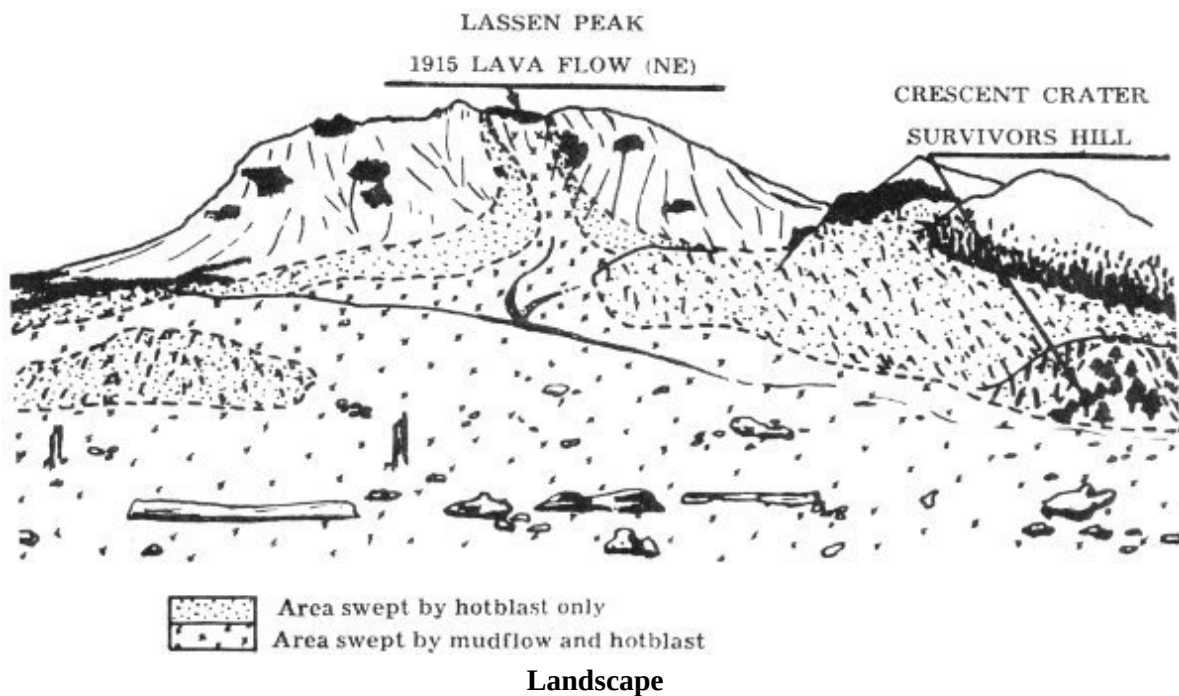
(0.4 mile)

43 HAT MOUNTAIN. This volcanic cone has an elevation of 7,695 feet, rising about 1,000 feet above Summit Lake near its south base.

(0.1 mile)

44 HEART OF THE DEVASTATED AREA. 10,457-foot Lassen Peak rises about 4,000 feet to the southwest. It is the largest plug volcano known. It was formed rapidly, being forced up as a stiff pasty mass of lava.

On May 30, 1914, without warning, Lassen Peak started a series of eruptions which lasted through 1917. The hundred eruptions during the first year were steam explosions which threw out ash, cinders and boulders, thus clearing out a new crater. On May 19, 1915, a black dacite lava flow welled up into this crater and spilled over to the southwest and northeast. At the southwest the lava descended about 1,000 feet on the Sacramento Valley side of the mountain, cooled and hardened. However, the lava coming down the northeast slope broke off in large pasty chunks, quickly melting a huge accumulation of snow. The resulting water, mixed with the large volume of fine and coarse debris from earlier eruptions formed a great mudflow which roared down the mountainside. Divided by Raker Peak, part of this mudflow went down Lost Creek, which the main road now follows northward. The rest went over the 100 foot rise in the east and down that Creek. It cut a swath through the forest which had been continuous across Lassen's lower and middle slopes. Fertile meadows were covered by as much as 20 feet of mudflow debris. The bark of trees on the edge of the mudflow was pounded and ripped off as high as 18 feet above the ground.



LASSEN PEAK
1915 LAVA FLOW (NE)

CRESCENT CRATER SURVIVORS HILL

Area swept by hotblast only

Area swept by mudflow and hotblast

Three days later (May 22, 1915) a great explosive eruption blasted out a new crater atop Lassen Peak, just west of the one which had been filled with, and sealed off, by the new lava. A portion of the explosive force was deflected downward. The resulting Hot Blast took the same, but wider, path the mudflow had taken. Trees left standing along the mudflow margins were uprooted or broken off by the blast and thrown down, all pointing away from Lassen Peak for a distance of 3 miles from the summit crater. Of the logs and snags which constitute an important part of this evidence, many have already rotted and weathered away since 1915. Some still show the abrasion and impact marks of mudflow and hotblast missiles.

The young trees growing in the Devastated Area have not been planted by man, but are natural reforestation. In National Parks nature takes its course. Man's meddling, even though well intentioned, often upsets the balance of nature, with resultant undesirable and unforeseen effects. 25

In summertime christine lupine (cream-colored) is abundant here. In the fall, rabbitbrush (yellow) is conspicuous.



RABBITBRUSH (yellow)



CHRISTINE LUPINE (pale yellow)

In late summer and fall, deer are numerous on the lower slopes of Lassen Peak. Some of the deer seen here are mule deer and its more common subspecies called the black-tailed deer, both recognized by their tails. The subspecies has a tail entirely black, and the mule deer has only the lower tip black with the remainder the color of its coat.

(0.7 mile)



MULE DEER MIGRATE TO THE EAST



BLACK TAILED DEER MIGRATE TO THE WEST

45 RAKER PEAK to the north is a 1,200-foot cone of andesite lava through which a stiff dome of dacite later welled up, the latter forming the cliffs. Logs felled by Lassen's hot blast of May 22, 1915, are also visible on its lower slope. The mountain was named Raker Peak in 1933 to perpetuate the memory of John E. Raker, Representative from California and author of the successful bill making Lassen a National Park in 1916.

26

Across the road to the west, on the left base of the slope, is a small patch of mature trees. The mound, called "SURVIVORS HILL," protected these trees from both the mudflow and the great Hot Blast. Survivors Hill also protected

a patch of forest on Raker Peak where a patch of old trees still stands unscathed, surrounded by down timber.

(0.4 mile)

46 OLD BOUNDARY SPRING lies in the grove of white-trunked quaking aspen trees just southwest and below the level of the Road. The spring provides excellent drinking water. Through this site once passed the original park boundary, hence the name.

Quaking aspens are sun-tolerant trees, members of the poplar family. In autumn, their leaves turn a golden-yellow, sometimes tinged with red. It is a short-lived tree and is often a pioneer in areas denuded by fire or other means. It provides shelter and water-retention, assisting in the establishment of young coniferous trees which eventually crowd out the aspens.

(0.5 mile)

47 An EMERGENCY TELEPHONE is on the east side of the Park Road.

(0.4 mile)

48 HOT ROCK is one of the many large boulders carried from the top of Lassen Peak. In the mudflow of May 19, 1915, these boulders were known to the local people as “hot rocks” because they retained their heat for several days. These which were buried in the mudflow material formed enough steam to cause funnel-shaped “eruption pits” in the mud overlying them. These pits can still be seen by careful searching in the upper portion of the Devastated Area.



The Lassen Park Road is closed in winter beyond the skiing area at [No. 3](#) in the Road Guide.
—NPS photo by R. C. Milne



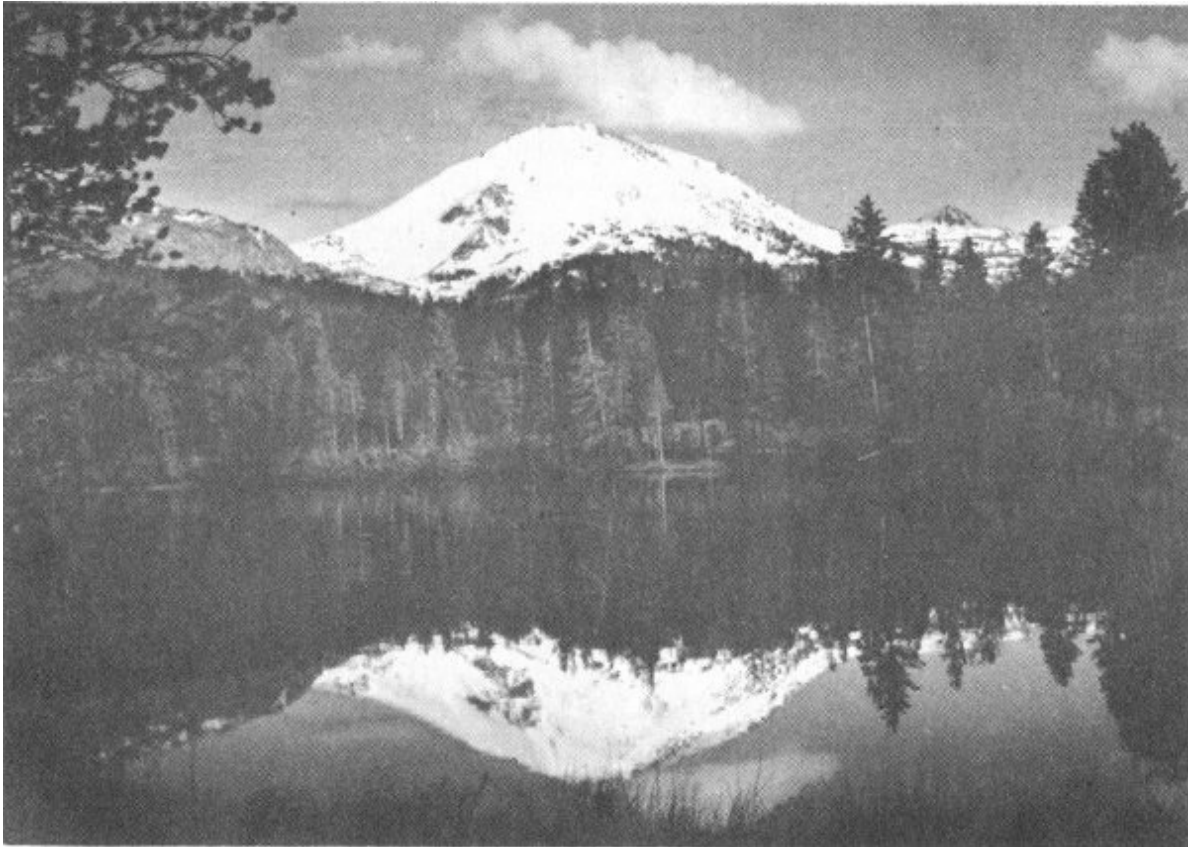
When the road is finally opened in early June, visitors drive through 20 ft. deep canyons of snow.

—NPS photo by R. C. Milne



Brokeoff Mountain, described at [No. 7](#) in the Road Guide, is part of the remnants of ancient Mt. Tehama.

—NPS photo by R. C. Milne



A good place to photograph Lassen Peak at sunset is at Reflection Lake, [No. 65](#) in the Road Guide.

—NPS photo by R. C. Milne

The numerous log and boulder jams were built up behind obstacles as the mudflow began to lose its tremendous size and momentum. The new forest growth is partly obscuring this feature, and before long all traces of the devastation will be obscured. 27

(0.4 mile)

49 LOST CREEK CROSSING. This stream has cut a trench not only through Lassen's recent mudflow, but through earlier ones, some of which were separated by rather long periods of time. Upstream can be seen two lower layers of tree stumps, still erect and in place, laid bare by the erosive action of Lost Creek.

This is also a good place to observe the activities of the water ouzels.

The Nobles Trail parallels the road a short distance to the east. It was in use until completion of the main road through the park in 1934.

(0.1 mile)

50 WEST BOUNDARY OF THE DEVASTATED AREA. The Great Hot Blast did not affect this area and only the effect of the mudflow of May 19, 1915, is visible. Recent growth of vegetation is healing the scars, but a change in soil, distribution of battered-down logs, barked tree trunks, and the change in the profile of the stream-cut bank across Lost Creek can still be seen.

(0.6 mile)

51 The LOST CREEK DIVERSION DITCH is below on the outside edge of the road. It was dug by Italian labor crews for the Shasta Power Company as a part of a water collection system for Battle Creek electrical development before this region became a National Park.

(0.1 mile)

52 On the inside bank of this turn, above the road, is a cross section of the LOST CREEK DIVERSION DITCH. The eruption of Lassen Peak on May 19, 1915, ended a long controversy between the power companies and the pioneer settlers regarding water rights by destroying, shortly after completion, the whole intake section at a point not far from here, upstream.

(0.9 mile)

53 In this area and on the slope above is an intermediate step in NATURAL REFORESTATION after a fire. The luxuriant growth of young white fir and jeffrey pine has been possible through soil stabilization and moisture retention accomplished by shrubs.

(0.5 mile)

54 A fine specimen of SUGAR PINE.

(0.3 mile)

55 LOST CREEK ORGANIZATIONAL CAMP is used only by organized groups. Reservations for this and other free group campgrounds in the park must be made in advance by writing to the Superintendent, Lassen Volcanic National Park, Mineral, Calif., 96063. Camping space in the other campgrounds is available on a first come first served basis.

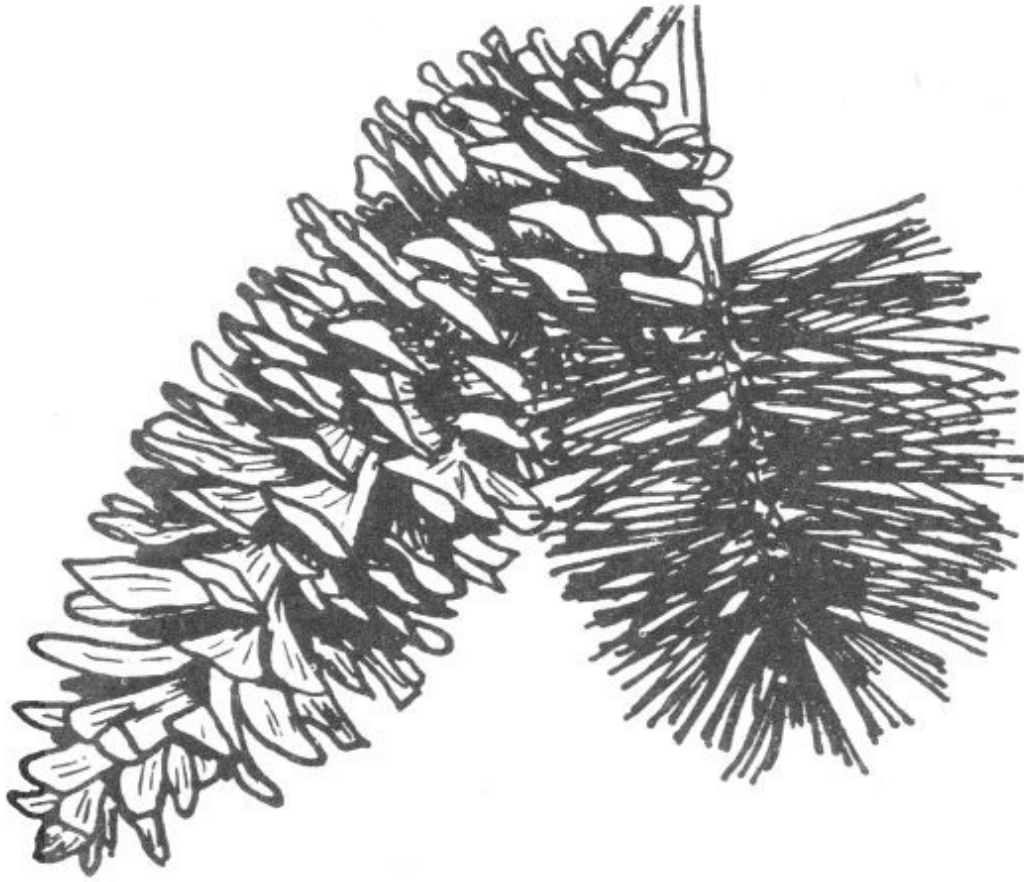
(0.2 mile)

56 A magnificent PONDEROSA PINE.

(0.2 mile)

57 This stretch of forest, especially on the east side of the road, may be called a “MICRO-CLIMATE.” The southwest exposure, in a sheltered depression, produces a warmer climate which supports white fir, incense cedar, ponderosa pine, sugar pine, shrubs and annuals characteristic of lower elevations. The green carpet covering the low rocks and parts of the forest floor is called squawcarpet ceanothus.

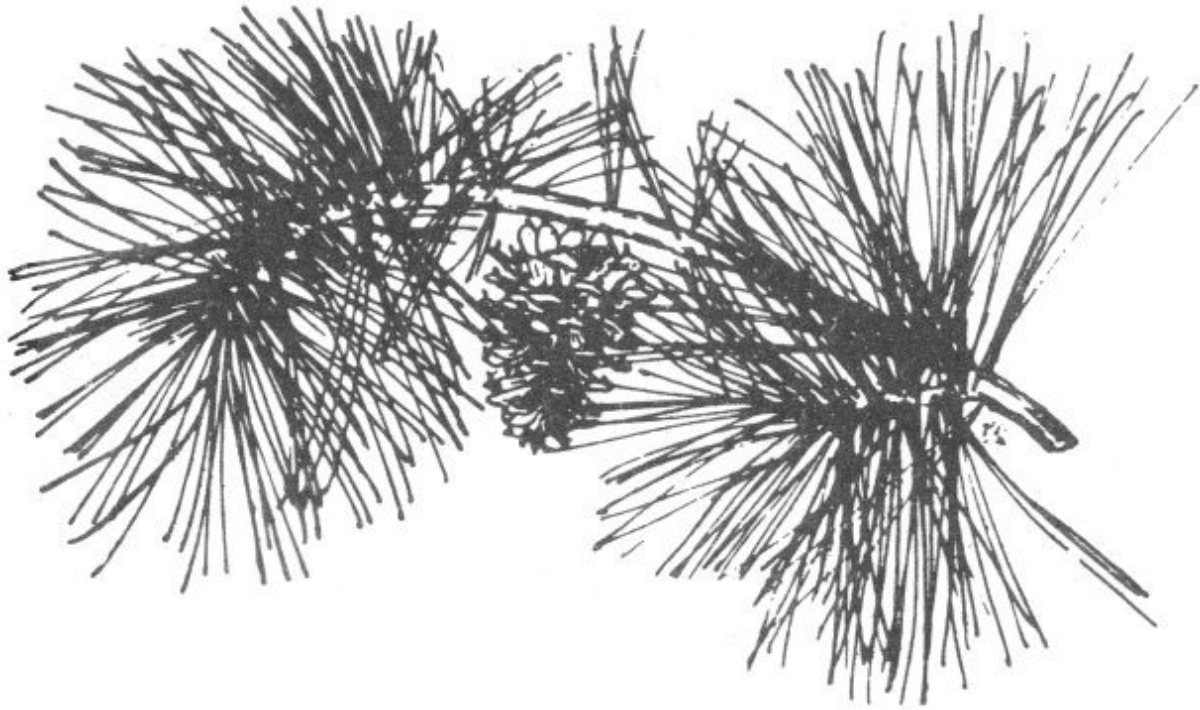
(0.3 mile)



SUGAR PINE (cone 12 inches)



INCENSE CEDAR (cone 1 inch)



PONDEROSA PINE (cone 3 inches)

58 On the eastern extension of the sharp curve is a loose **ROCKY TALUS SLOPE** composed of fine-grained, dark grey boulders known as basalt. Often basaltic lavas are very porous because they are typically gas-rich when molten, and many bubble holes result as the escaping gases are trapped during cooling. This lava is relatively low in silica and high in iron, hence it is dark in color and heavy in weight.

(0.6 mile)

59 **PROSPECT PEAKS** are Hawaiian or shield volcanos made of countless thin basaltic lava flows from a central vent, producing shield-shaped outlines. The sharp peaks on top of these two mountains are small cinder cones. Badger Mountain is a plug volcano like Chaos Crags or Lassen Peak. West Peak is outside the park and on it is a Forest Service fire look-out station. To the north and northeast are typical virgin forests mantling a sea of minor volcanic peaks.



Landscape

LATOUR BUTTE
 MAGEE PEAK
 SUGAR LOAF PEAK
 HAT CREEK RIM
 HAT CREEK FAULT
 WEST PROSPECT PEAK
 PROSPECT PEAK
 BADGER MT.

The HAT CREEK FAULT SCARP, outside the park, is a great crack in the earth where an enormous block to the east, known as the Hat Creek Rim, rose. The scarp, or bluff, runs about 25 miles north. This is the beginning of the so-called basin range structure of Nevada which is largely composed of such variously tilted fault blocks.

(0.6 mile)

60 NOBLES PASS, elevation about 6,000 feet, was discovered by William H. Nobles. In 1852 he offered to reveal the route to business men of Shasta City, then the leading settlement of Northern California, for \$2,000. The new route, the most popular into this region, started at Black Rock, Nevada, and entered the northeast corner of the park at Butte Lake. This Nobles, or Old Emigrant, Trail is still visible to the northeast through the chaparral. From this point, it went west along the south base of Table Mountain and left the park near the highway "Y" beyond Manzanita Lake. Numerous authentic pioneer relics have been found along the route.

The chaparral brush formation here is composed of greenleaf manzanita, snowbrush, ceanothus, and bitter cherry. Establishment of the chaparral is

often the first step in the natural reforestation of an area swept by forest fires.

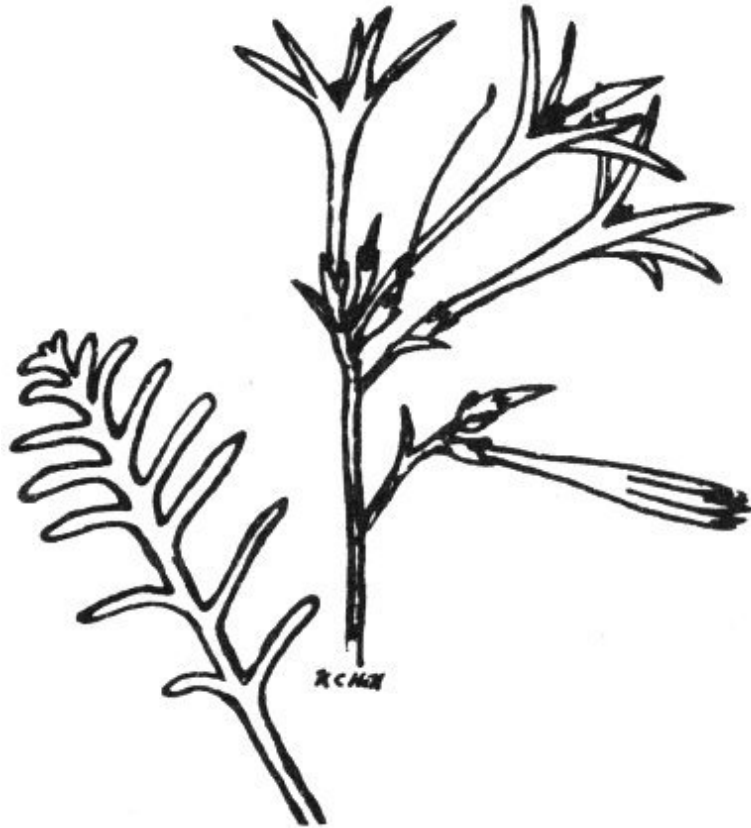
(0.4 mile)

61 This is one of the largest WHITE FIR trees in the United States. It is 168 feet high and has a circumference of 20 feet, 7 inches.



WHITE FIR

This vicinity is known as SUNFLOWER FLAT due to the abundance of the flower-like plant called wooly wyethia which blossoms in mid-summer. Other flowers to be seen are: skyrocket gilia, California stickweed, Nuttall larkspur, and Pacific monardella. A few brilliant red snowplants are often to be found here up into mid-summer. Remember, no flower picking is allowed in our National Parks.



SKYROCKET GILIA (red)



CALIFORNIA STICKSEED (white)



NUTTALL LARKSPUR (blue)



PACIFIC MONARDELLA (white to lavender)

(0.2 mile)

62 The sudden change in topography indicates the east edge of CHAOS JUMBLES ... the huge landslide from nearby Chaos Crag.

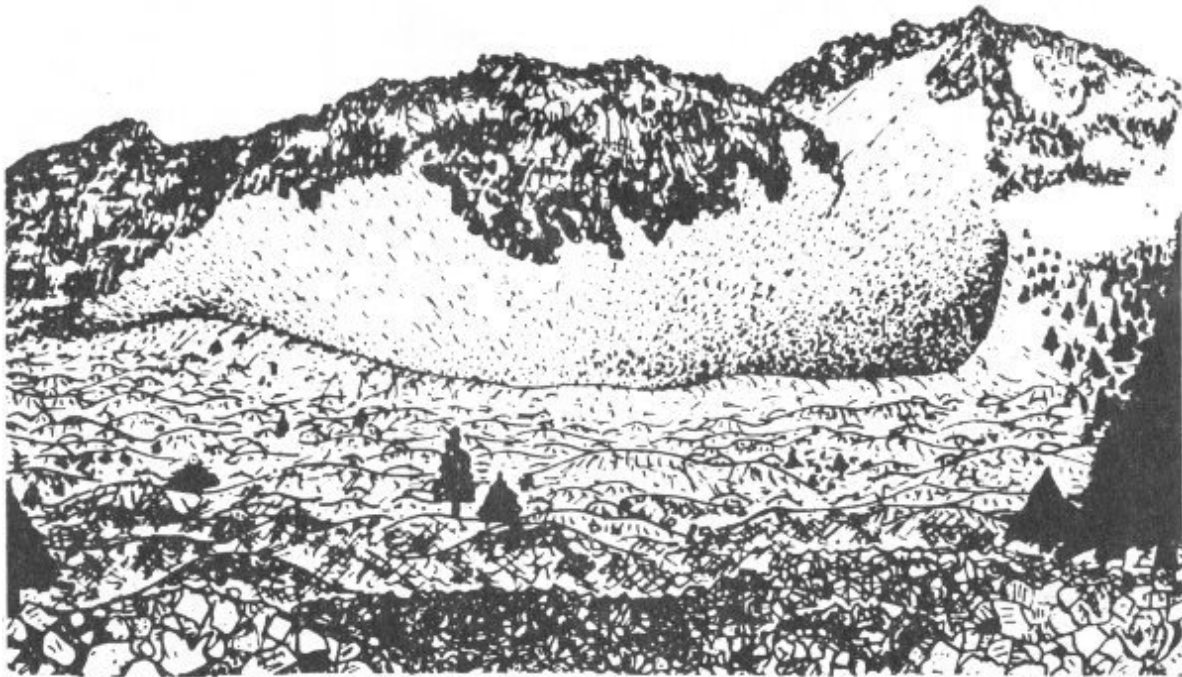
This upper edge of the hummocky rock debris is thin, and lies on a relatively flat undersurface. As a result, the forest trees grow normally here, in contrast to the dwarf forest to be found on the main mass of thick, sloping, and very porous Jumbles. 32

The pale yellow-green, moss-like material on the tree trunks is called staghorn lichen. It is not a moss, but is a union of two plants growing together; a fungus, giving it form and body, and an alga, growing inside and manufacturing food for both from the air and sunshine. Lichens do not in any way harm the trees, and grow as well on dead material.

(0.6 mile)

63 The pink mountain mass to the south is CHAOS CRAGS. It is composed of three plug volcanos of a lava known as dacite. This material was rapidly pushed up in a molten state, much as one would squeeze paste from a tube. As the lava plugs cooled and hardened, steam explosions took place from the base of the cliff, undermining it and causing at least three tremendous avalanches estimated as occurring about 1700 years ago, 700 years ago and the most recent in about the year 1690. Due to the small amount of soil and porous rock, many of the stunted trees in this Dwarf Forest are over 200 years old.

(1.1 mile)



Landscape

64 VISITOR CENTER, LODGE and CAMPGROUND, elevation 5,950 feet, is the center of visitor facilities and services.

33

The Visitor Center is open daily. Free naturalist-conducted hikes, caravans, and nature walks originate here. Informal campfire programs are presented

nightly in the Manzanita Lake Campground Amphitheater from June through mid-September.

The LILY POND SELF-GUIDING NATURE TRAIL, an easy 1-mile loop route, starts just across the road from the Visitor Center. It is unique in having ten species of cone-bearing trees and three species of broad-leaved trees along its short and nearly level route.



GOLDEN-MANTLED GROUND SQUIRREL · CHIPMUNK

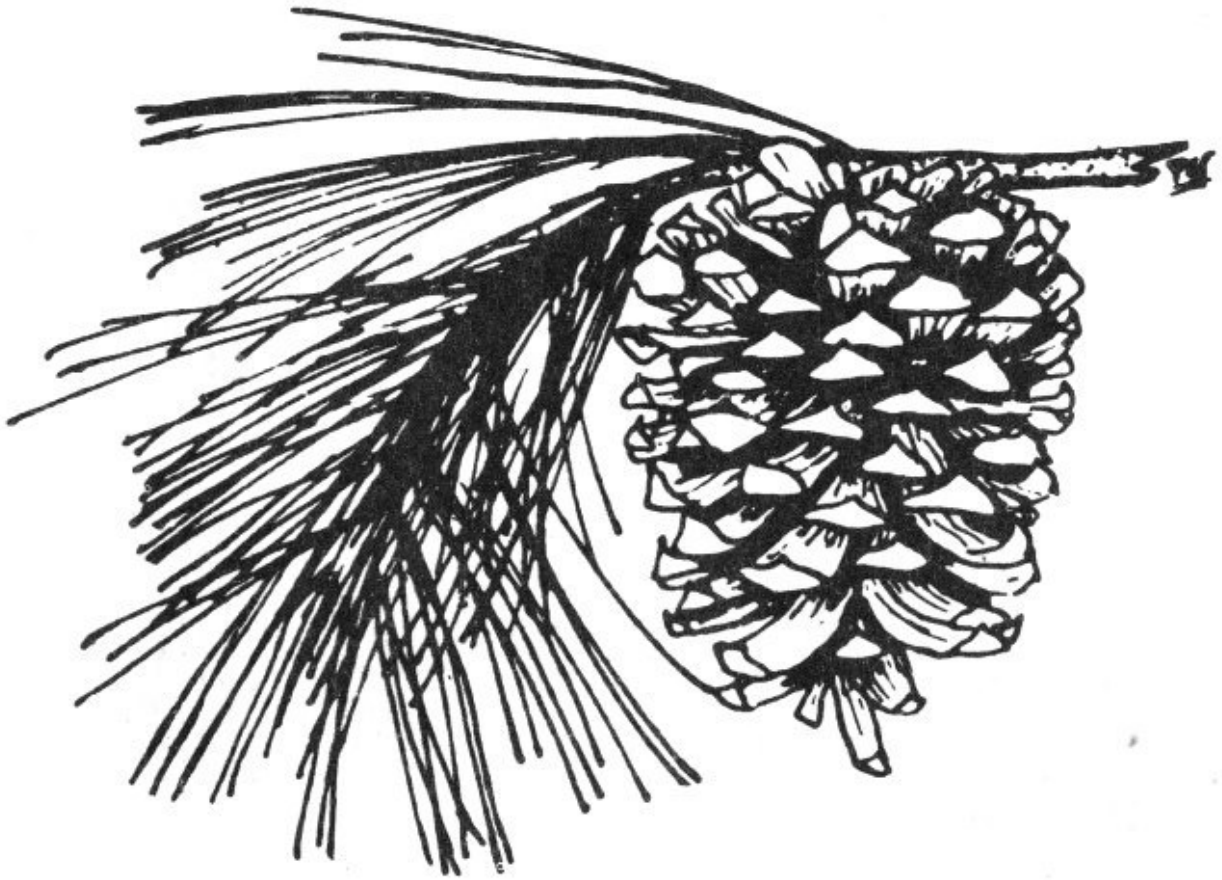
Abundant wildlife, including deer and a large variety of birds, is found in the Manzanita-Reflection Lakes area. The most common rodents are the golden-mantled ground squirrel and the chipmunk.

The Loomis Museum, with a tract of 40 acres, was given to the Federal Government in 1927 by Mr. and Mrs. Benjamin F. Loomis in memory of their daughter, Mae. On this land are now located the public lodge and dining room, curio store, service station, grocery store and postoffice.



The National Park Service emblem pictured here was adopted in 1953. Many visitors confuse the Park Service (Department of the Interior) with the Forest Service (Department of Agriculture). Briefly, the National Park Service preserves the National Parks and Monuments in an unchanged state, while the U. S. Forest Service operates National Forests on a sustaining yield basis. 34

(0.1 mile)



JEFFREY PINE (cone about 5 inches)



SNOWPLANT (red)



COW LILY (yellow)

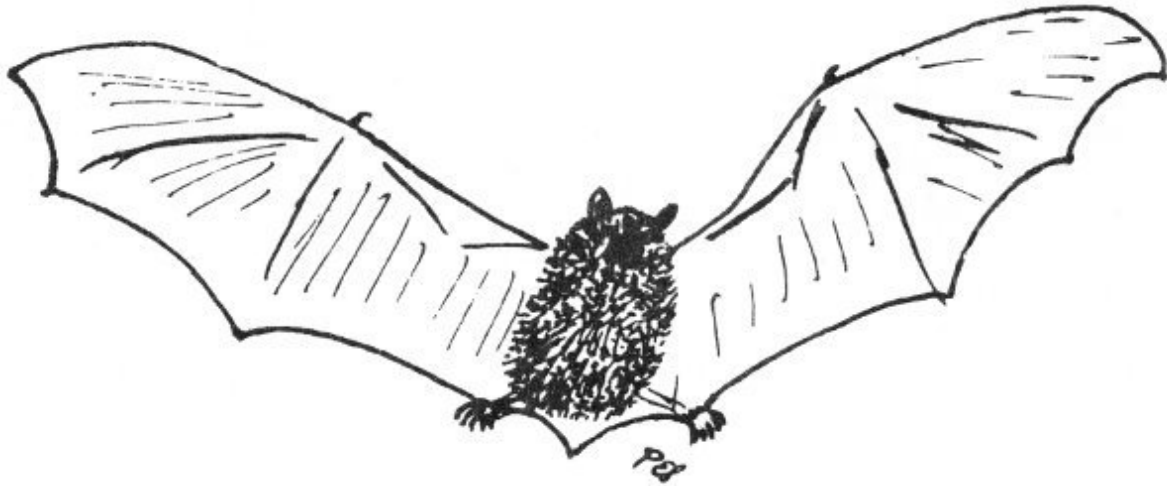
65 REFLECTION LAKE, elevation 5,880 feet. A few hundred feet ahead is the road to the Reflection Lake Picnic Ground. There tables, fireplaces, and rest rooms are available for free visitor use.

35

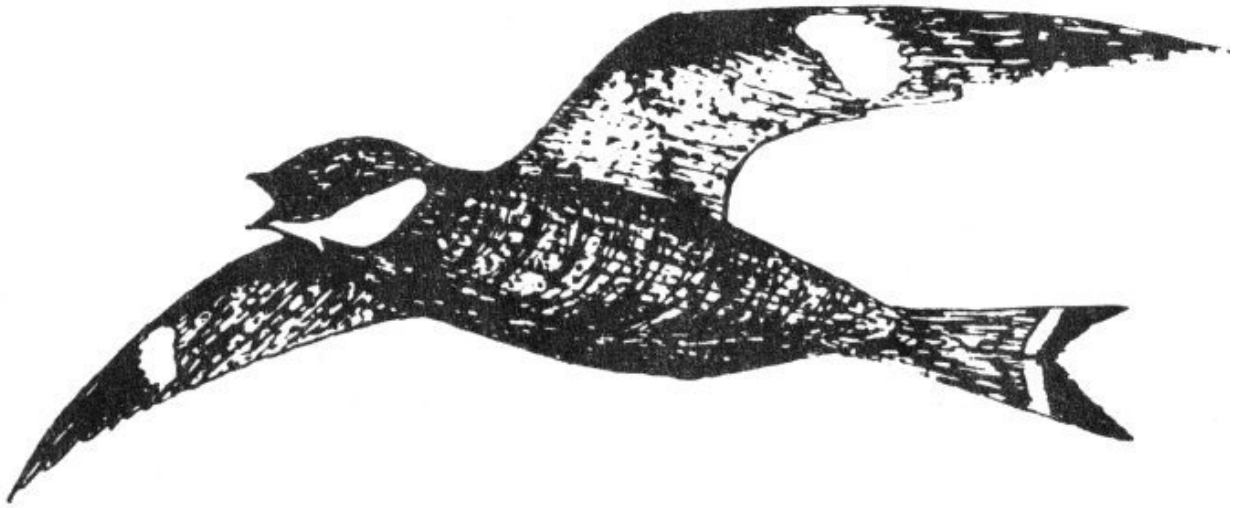
Reflection Lake gets its name from the fine reflections of Lassen Peak and Chaos Crag to be seen in it from near the picnic ground. It is an excellent place for photography, especially in the afternoon or early evening.

On summer evenings nighthawks can be seen cruising over the lake, emitting their unique harsh cries and their startling “whoomm” sound in flight. Bats on the wing also are seen in the evenings feeding on insects.

(0.2 mile)



BROWN BAT



NIGHTHAWK

66 On the down hill side of the road is MANZANITA LAKE, elevation 5,845 feet. This lake is young, formed by the Chaos Jumbles landslide. The water level was raised a few feet in 1912 by the Northern California Power Company, for electric power development purposes by the construction of a small earth dam at the lake's southwest extremity.

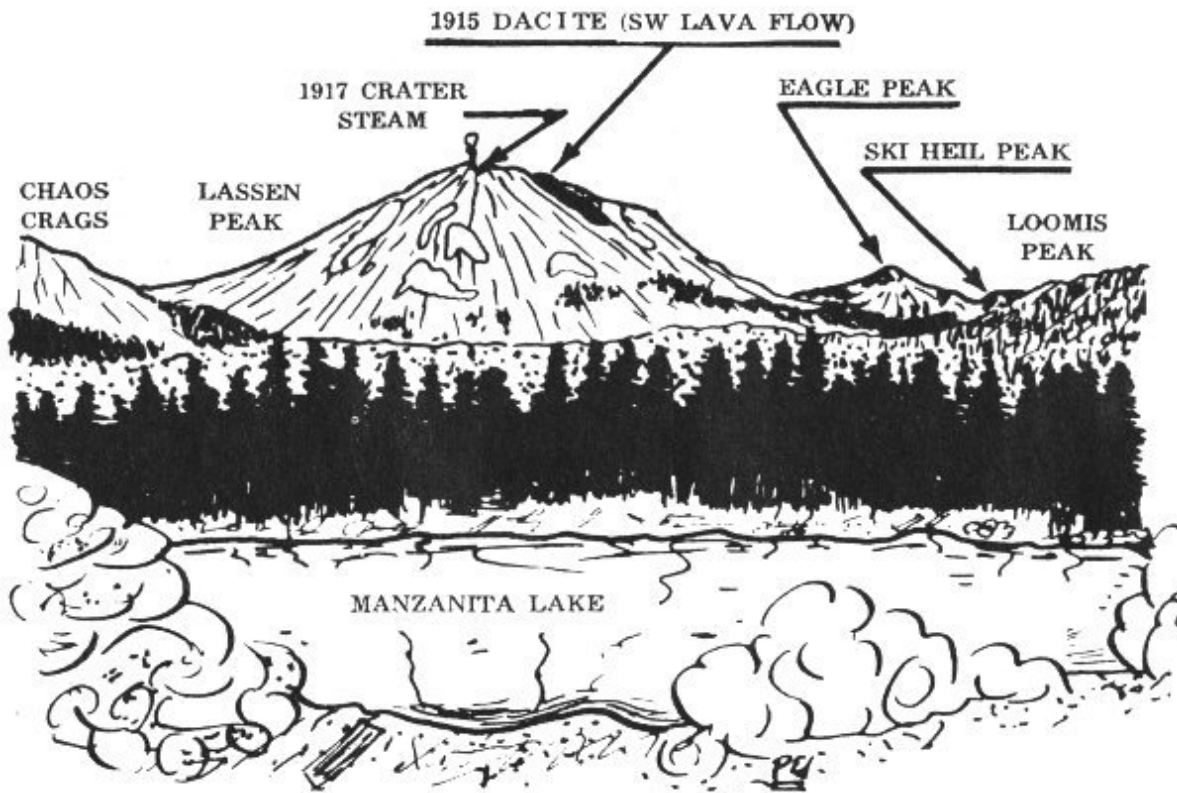
Like most of Lassen's lakes, it is well stocked with trout. In the fall large numbers of waterfowl, including Canada geese, stop here en route to their wintering grounds.

The lake derived its name from the abundance of greenleaf manzanita on its shores. This shrub can be seen across the lake to the south.

(0.2 mile)

67 MANZANITA LAKE ENTRANCE STATION. Vehicle entry permits and information may be procured daily during the summer season.

Lovely views of Lassen Peak are to be had along the Lake Trail just a few feet from the Entrance Station at this end of the lake. It is a fine area for taking photographs, especially in the afternoon and early evening. Lassen Peak towers 4,625 feet above Manzanita Lake to an elevation of 10,457 feet. The summit notch in the 1917 crater, and the black tongue extending down toward the right is the youngest rock in contiguous United States, pouring out as molten, redhot dacite lava on the night of May 19, 1915.



Landscape

1915 DACITE (SW LAVA FLOW)
1917 CRATER STEAM
EAGLE PEAK
SKI HEIL PEAK
CHAOS CRAGS
LASSEN PEAK
LOOMIS PEAK
MANZANITA LAKE

AUTO TRIPS TO OTHER POINTS OF INTEREST

1. BIG SPRING, SUBWAY CAVE (lava tube), and BURNEY FALLS STATE PARK, 8, 14, and 41 miles respectively north of Manzanita Lake on Highway No. 89.
2. BUTTE LAKE CAMPGROUND and 1½-mile hike over the Old Emigrant Trail to CINDER CONE (Self-Guiding Nature Trail). A 30-mile drive from Manzanita Lake; proceed north on Highway No. 44 for nearly 15 miles, east, and later south, following the signs 15 miles to Butte Lake.
3. WARNER VALLEY and DRAKESBAD: South and east on Highways No. 89 and No. 36 to Chester, then north and west on an oiled road 18 miles to Drakesbad. (Self-guiding Nature Trail).
4. JUNIPER and HORSESHOE LAKES: via dirt roads about 18 miles north from Chester.



The Loomis Museum Association is a non-profit organization which is dedicated to the accumulation and dissemination of information concerning the human and natural history of Lassen Volcanic National Park. The following publications are available at the Manzanita Lake Visitor Center during the summer and at Park Headquarters in Mineral all year. Mail orders should be addressed to Loomis Museum Association, Mineral, Calif. 96063. Price lists are available on request.

38

LASSEN—HIS LIFE AND LEGACY—
GEOLOGY OF LASSEN'S LANDSCAPE—
PICTORIAL HISTORY OF THE LASSEN VOLCANO—
ROAD GUIDE TO LASSEN VOLCANIC NATIONAL PARK—
INDIANS OF LASSEN VOLCANIC NATIONAL PARK AND VICINITY
—
BIRDS OF LASSEN VOLCANIC NATIONAL PARK—
FLOWERS OF LASSEN—
FIELD GUIDE TO THE TREES AND SHRUBS OF LASSEN VOLCANIC
NATIONAL PARK—
AMPHIBIANS AND REPTILES OF LASSEN VOLCANIC NATIONAL
PARK—



39

**ASK
THE MAN IN THE**

NATIONAL PARK SERVICE UNIFORM



He'll be glad to help you!

BE PROUD OF THIS PARK

As a citizen of the United States it belongs to you.

Keep it unspoiled for your next visit and for future generations by helping to:

Prevent forest fires.

Protect the flowers, the animal life,
and the rock and mineral formations.

Keep it clean.



Transcriber's Note

- A few obvious typographical errors were corrected.

*** END OF THE PROJECT GUTENBERG EBOOK ROAD GUIDE TO
LASSEN VOLCANIC NATIONAL PARK ***

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